

# Preface

2019年5月14日 12:09

## L1: Bisection & Fixed Point Iteration

2019年5月14日 12:10

### 1. Introduction

#### a. Textbook

For appl: Y & M; Y

### 2. Introduction

#### a. Tips

##### i. Analytic

Solved with symble, without iteration

##### ii. Systematic

Include factor in solution

##### iii. Type of answer

Sometimes a numerical answer is more useful

#### b. Problem with brutal force numerical method

When calculation is expensive

When continuity does not hold

When existence and uniqueness does not hold

When we are not given a finite range

#### c. Requirement for Well-posed question

##### i. Requirements

1) Continuity

2) Differentiability

3) Smoothness  $C^k$

4) Lipschitz continuity

##### ii. Lipschitz continuity

###### 1) Definition

$$\left( \exists c \in \mathbb{R} \forall x_1, x_2 |f(x_1) - f(x_2)| \leq c|x_1 - x_2| \right) \Rightarrow \text{Lipschitz continuity}$$

###### 2) Description

Average rate of change is bounded

###### 3) Average and Instantaneous

Instantaneous rate of change bounded everywhere  $\Rightarrow$  average rate of change bounded  $\Rightarrow$  Lipschitz continuity

###### iii. Relation

$$\underbrace{\text{Continuously Differentiable}} \subset \underbrace{\text{Differentiable}} \subset \underbrace{\text{Lipschitz Continuous}} \subset \underbrace{\text{Continuous}}$$

#### iv. Why requirement matters

Requirement is the range of appliance; according to the range, we apply different suitable method.

### d. Method

#### i. Bisection

##### 1) Condition:

✓ (1) Continuity & Proper initial interval

##### 2) Method

Keep reducing the interval  $(x_1, x_2)$  where  $f(x_1)f(x_2) < 0$

##### ✓ 3) Theorem behind: Intermediate Value Theorem

$$(f: [a, b] \rightarrow \mathbb{R} \text{ is continuous}) \Rightarrow \left( \forall k \in (f(a), f(b)) \exists x \in (a, b) f(x) = k \right)$$

##### 4) Convergence Rate

###### a) Absolute Error

$$\text{for } [a_k, b_k] \quad E_k = \left| \frac{b_k - a_k}{2} - x^* \right| \leq \frac{1}{2} |b_k - a_k|$$

###### b) Convergence Rate

$$\frac{E_{k+1}}{E_k} \leq \frac{1}{2}$$

\*think about why

##### 5) Problem

a) Cannot assure that all solutions are found

b) Need an proper initial interval

#### ii. Fixed point Iteration

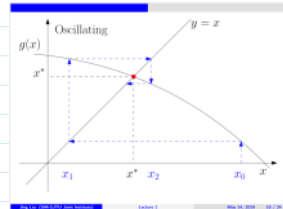
##### ✓ 1) Condition: (4) Lipschitz Continuity

$f(x) = g(x) - x$ , where  $g(x)$  has Lipschitz constant  $c \in (0, 1)$

##### 2) Method

$$x_{k+1} = g(x_k)$$

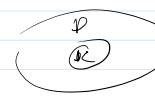
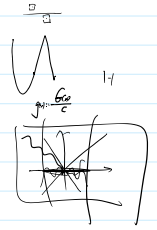
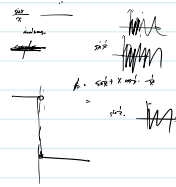
##### 3) Scheme



##### 4) Notes

Extension will be discussed in assignment

$$\frac{d}{dr} \frac{1}{r^2} = -\frac{2}{r^3}$$



$$\left[ \frac{c}{k} \right] \rightarrow \frac{1}{k}$$

## L2: Newton's & Secant Method

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### 1. Root Finding

#### a. Fixed point Iteration

##### i. Condition

$f(x) = g(x) - x$ , where  $g(x)$  has Lipschitz constant  $c \in (0,1)$

##### ii. Method

$$x_{k+1} = g(x_k)$$

##### iii. Condition for quadratic convergence

$g(x)$  need to be twice differentiable

##### iv. Proof of quadratic convergence

Taylor expansion

#### b. Newton's Method

##### i. Method

$$x_{k+1} := x_k - \frac{f(x_k)}{f'(x_k)}$$

##### ii. Requirements I guess

1) Once differentiable;

2) First derivative should always be positive or negative

3) Choice of  $x$  should be close enough

How close? We cannot say

##### iii. Theorem

##### 1) Theorem

$$\left( f \in C^2[a,b], \exists x^* \in [a,b] f(x^*) = 0 \right) \Rightarrow \left( f'(x^*) \neq 0 \Rightarrow \left( \exists \delta > 0 \quad \forall x_0 \in [x^* - \delta, x^* + \delta] \left\{ x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \right\}_{n=0}^{\infty} \xrightarrow{\text{quadratically}} x^* \right) \right)$$

##### 2) Notes

It not only converges, but also converges quadratically

##### 3) Proof of Convergence

Transfer to  $g(x) = x - f(x)/f'(x)$ , so as to use fixed-point method

We need to choose  $|g'(x)| < 1$  for initial condition

##### 4) Sufficient condition for quadratic convergence:

$f$  three times differentiable

- Clearly if additionally  $f$  is three times differentiable, and has nonzero slope

$$f'(x) \neq 0 \quad \text{for all } x.$$

then Newton's method surely has quadratic convergence and behave nicely

$$g'' = \frac{(f')^2 \cdot f'' - 2f \cdot (f'')^2 + f \cdot f' \cdot f'''}{(f')^3}$$

once  $x_0$  is chosen correctly.

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Lecture 2

May 16, 2019

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##### 5) Proof of quadratic convergence:

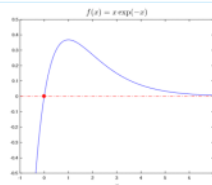
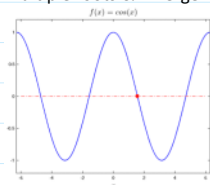
Second order Taylor expansion

##### iv. How to choose $x_0$

$$x_0 \in I \quad \forall x \in I \quad \left| \frac{f(x) \cdot f''(x)}{f'(x)^2} \right| < 1$$

##### v. Failure cases

Multiple roots & Divergence



#### c. Secant Method

##### i. Introduction

Newton's method is too expensive: need evaluation of  $f$  and  $f'$  for one evaluation of  $x_k$

##### ii. Requirement

???

##### iii. Method

Substitute  $\frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}$  for  $f'(x_k)$  in Newton's Method

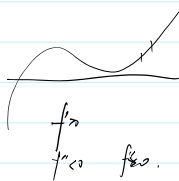
##### iv. Equation

$$x_{k+1} = x_k - \frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})} \cdot f(x_k)$$

### 2. Optimization

#### a. Basics

Global extrema and local extrema



# L3: Optimization

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## 1. Optimization: Finding Extrema

### a. Condition for local extrema under differentiability

#### i. Introduction

? In which arrow are we looking for necessary and sufficient condition?

We are looking for the link of the red arrow

#### ii. First Order

##### 1) Fermat Theorem:

$$\left( \begin{array}{c} \square \\ \text{once differentiable} \\ \square \end{array} \right) \Rightarrow \left( \begin{array}{c} \square \\ [\text{local minimum at } x^*] \Rightarrow [f'(x^*) = 0] \\ \square \end{array} \right)$$

##### 2) First order necessary condition

- For a smooth function  $f(x)$  defined on a closed bounded interval  $[a, b]$ ,

If the boundary point  $a$  is a global minimiser, then  $f'(a^+) \geq 0$ .

If an interior point  $x^* \in (a, b)$  is a global minimiser, then  $f'(x^*) = 0$ .

If the boundary point  $b$  is a global minimiser, then  $f'(b^-) \leq 0$ .

#### iii. Second Order

##### 1) Theorem

$$\left( \begin{array}{c} \square \\ \text{twice continuously} \\ \text{differentiable} \\ \text{in open neighbourhood} \\ \square \end{array} \right) \Rightarrow \left( \begin{array}{c} \square \\ \left\{ \begin{array}{l} f'(x^*) = 0 \\ f''(x^*) > 0 \end{array} \right\} \Rightarrow [\text{local minimum at } x^*] \\ \square \end{array} \right)$$

##### 2) Whether red arrow is necessary?

We are questioning whether  $f''(x^*) > 0$  is necessary

If we add the case of  $f''(x^*) = 0$ , the red arrow becomes necessary but not sufficient

### b. How to solve for extrema

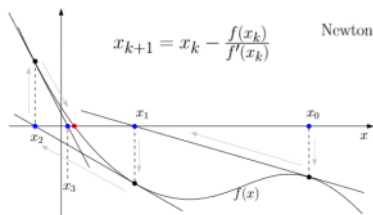
#### i. Newton's Root finding for $f'(x) = 0$

##### 1) Equation

$$x_p = x_k - \frac{f'(x_k)}{f''(x_k)}$$

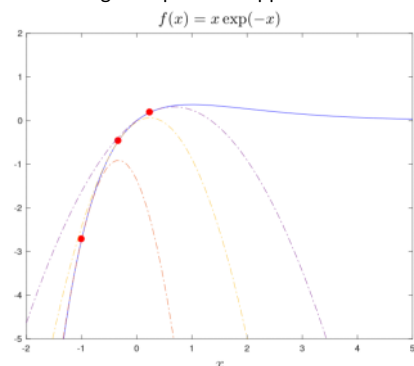
##### 2) Understanding 1

We are using Newton's method to find the root for  $f'$



##### 3) Understanding 2

We are doing local parabola approximation



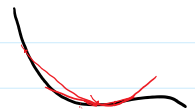
##### 4) Requirement

Twice differentiability (need not be twice continuously differentiable)

#### ii. Successive parabolic interpretation

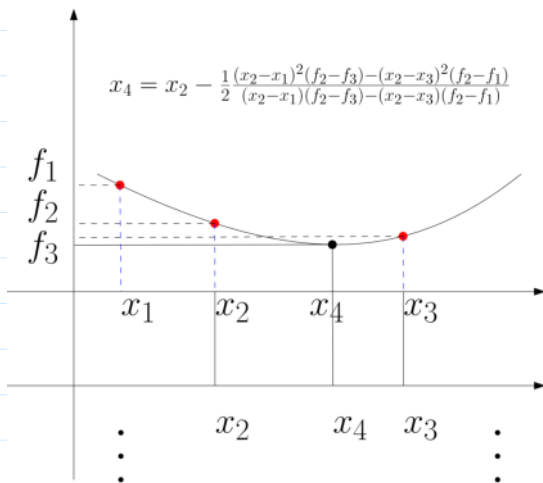
##### 1) Method

Non-local parabola approximation



from before we saw a before a root





## 2) Requirement

### 3) Problem: Cannot get start

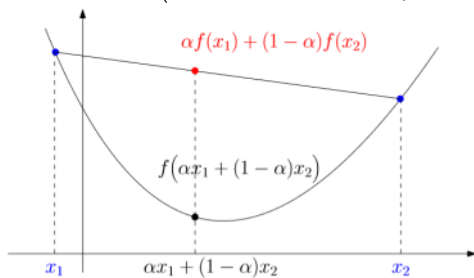
Initial  $x_2$  shouldn't be at the minimum of parabola

## c. Without Differentiability

### i. Use Convex to replace second differentiability

#### 1) Definition

$$\left( \begin{array}{c} \square \\ f: \mathbb{R}^n \rightarrow \mathbb{R} \\ \square \end{array} \right) \Rightarrow \left( \begin{array}{c} \forall x_1, x_2 \in \mathbb{R}^n \quad \forall \alpha \in [0,1] \quad f(\alpha x_1 + (1-\alpha)x_2) \leq \alpha f(x_1) + (1-\alpha)f(x_2) \\ \Downarrow \\ f \text{ is convex} \end{array} \right)$$



## 2) Theorem

A local minimum of a convex function is necessarily a global minimum

### 3) Theorem: Sufficient and Necessary first order condition

$$\left( \begin{array}{c} \square \\ f: \mathbb{R} \rightarrow \mathbb{R} \\ \text{convex} \\ \text{differentiable} \end{array} \right) \Rightarrow \left( \begin{array}{c} \square \\ \text{global minimum at } x^* \Leftrightarrow f'(x^*) = 0 \\ \square \end{array} \right)$$

## ii. Golden Section Search: Without differentiability

### 1) Definition of unimodal

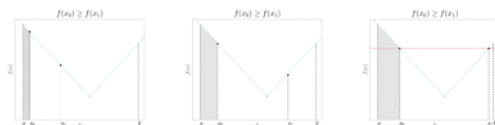
$$\left( \begin{array}{c} \square \\ f: \mathbb{R} \rightarrow \mathbb{R} \\ \square \end{array} \right) \Rightarrow \left( \begin{array}{c} f \text{ is unimodal on } \mathcal{I} = [a, b] \\ \Updownarrow \\ \exists_{x^* \in \mathcal{I}} \text{ strictly decreasing for } [a, x^*], \text{ strictly increasing for } [x^*, b] \end{array} \right)$$

## 2) Understanding

Quasi-convex

## 3) Method

- It is clear that there are three possibilities of  $f(x_0) \geq f(x_1)$  where  $x_1 > x_0$ :



all three possibilities have the minimum outside  $[a, x_0]$ .

- Note the the minimum cannot be in the interval  $[a, x_0]$ , so we conclude that

$$x^* \in (x_0, b)$$

- Similarly, if  $f(x_0) \leq f(x_1)$ , we can conclude that

$$x^* \in (a, x_1)$$

- Because of those two conclusions, we can always "squeeze" the interval

$$\mathcal{I} = [a, b]$$

after two evaluations of  $f$  at two interior points of  $\mathcal{I}$ .

- Evaluating  $f$  again at two interior points of the resulting interval

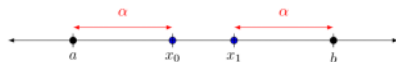
$$(x_0, b) \quad \text{or} \quad (a, x_1)$$

will squeeze the interval further, so we have tighter bounds for  $x^*$  each time.

Q: How shall we choose the interior points each time?

- Since we have no information on where  $x^*$  is more likely to be, so we choose to avoid arbitrary preference, e.g., the first two interior points are chosen so

$$x_0 - a = b - x_1 = \alpha$$



Q: How shall we choose  $\alpha$ ?

#### 4) How to choose $\alpha$

What do we want by choosing  $\alpha$

d. '

## L4: Gaussian Elimination and Back Substitution

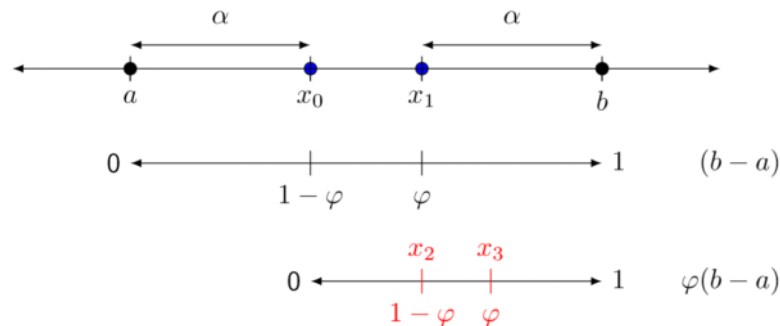
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### 1. Optimization

#### a. Golden Section Method

##### i. Scheme

- Let us suppose  $f(x_0) \geq f(x_1)$ , that is,  $[a, x_0]$  happens to be discarded, then



##### ii. How much to dig in

Dig in  $(1 - \varphi)$  part of the interval from both sides. To enhance reuse of evaluation, we choose  $\varphi = \frac{-1+\sqrt{5}}{2}$

##### iii. Characteristic

- 1) Converge unconditionally and linearly
- 2) No requiring differentiability

### 2. Linear System

#### a. System & Matrix

##### i. Operation: $E_{i,j}, E_{(\alpha)i}, E_{(\alpha)i,j}$

Type I : Interchange two rows  
 Type II : Multiply a row by a nonzero constant  
 Type III : Add a constant times one row to another

$$\begin{pmatrix} E_{i,j} \\ E_{(\alpha)i} \\ E_{(\alpha)i,j} \end{pmatrix}$$

##### ii. Gauss's Elimination Back Substitution

Eliminate all factors beneath main diagonal into row echelon form

$$\left[ \begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 0 & -\frac{1}{2} & -\frac{3}{2} \end{array} \right]$$

##### iii. Algorithm

## Algorithm 1: NAIVE GAUSSIAN ELIMINATION WITH BACK SUBS

**Input** : matrix  $[a_{ij}]$  of  $n \times n$  and matrix  $[b_{ij}]$  of  $n \times m$

**Output** : solution matrix  $[x_{ij}]$  of  $n \times m$

```

1 Function GaussNaive( $[a_{ij}]$ ,  $[b_{ij}]$ ):
2   for  $k \leftarrow 1$  to  $n - 1$  do
3     for  $i \leftarrow k + 1$  to  $n$  do
4        $m \leftarrow a_{ik}/a_{kk}$  ;                                /* Multiplier */
5        $a_{ik} \leftarrow 0$  ;                                    /* below pivot */
6       for  $j \leftarrow k + 1$  to  $n$  do
7          $a_{ij} \leftarrow a_{ij} - m \cdot a_{kj}$  ;                /* Type III operation */
8       end for
9       for  $j \leftarrow 1$  to  $m$  do
10         $b_{ij} \leftarrow b_{ij} - m \cdot b_{kj}$  ;                /* Type III operation */
11      end for
12    end for
13  end for
14  /* The loop above is the gaussian elimination, and the loop below is the back substitution */
15  for  $k \leftarrow 1$  to  $m$  do
16     $x_{nk} \leftarrow b_{nk}/a_{nn}$  ;                                /* Solution to the last variable */
17    for  $i \leftarrow n - 1$  to  $1$  step  $-1$  do
18       $z \leftarrow b_{ik}$  ;                                    /* the right-hand side */
19      for  $j \leftarrow i + 1$  to  $n$  do
20         $z \leftarrow z - a_{ij} \cdot x_{jk}$  ;                    /* one less left-hand term */
21      end for
22       $x_{ik} \leftarrow z/a_{ii}$  ;                                /* Solution to the variable above */
23    end for
24  end for
25  return  $[x_{ij}]$  ;                                          /* solutions */

```

### iv. Problem

- 1) Assuming Existence & Uniqueness of solution
- 2) Assuming none-zero pivot
- 3) Accumulation of float value errors

| Significant figures | $x_2$     | % Error in $x_2$ | $x_1$      | % Error in $x_1$ |
|---------------------|-----------|------------------|------------|------------------|
| 7                   | 0.6666667 | 0.000005         | 0.3330000  | 0.1              |
| 6                   | 0.6666670 | 0.00005          | 0.3300000  | 1                |
| 5                   | 0.6666700 | 0.0005           | 0.3000000  | 10               |
| 4                   | 0.6667000 | 0.005            | 0.0000000  | 100              |
| 3                   | 0.6670000 | 0.05             | -3.0000000 | 1000             |

To avoid this, we can choose pivot by swapping rows

### v. Pivoting

- 1) Partial Pivoting  
Getting small elements out of pivot
- 2) Scaled Partial Pivoting  
Getting relative small term out of pivot

### b. Determinant

#### i. Algorithm of $n \times n$ determinant

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 1 \\ 6 & -3 & 4 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & 3 \\ 0 & 0 & -5 \\ 0 & -6 & -5 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & 3 \\ 0 & -6 & -5 \\ 0 & 0 & -5 \end{bmatrix} = \mathbf{U}$$

#### ii. Operation

$$\det(\mathbf{E}_{i,j}\mathbf{A}) = -\det(\mathbf{A})$$

$$\det(\mathbf{E}_{(\alpha)i}\mathbf{A}) = \alpha \det(\mathbf{A})$$

$$\det(\mathbf{E}_{(\alpha)i,j}\mathbf{A}) = \det(\mathbf{A})$$

# L5: Multiplication & Determinant

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1. '

## a. Matrix Multiplication

### i. Understanding of Multiplication

#### 1) By element

Traditional Method

#### 2) By column



$$\begin{bmatrix} 6 & -3 & 4 \\ 0 & 0 & -5 \\ 6 & -3 & 4 \end{bmatrix} \quad \begin{bmatrix} 2 & 1 & 3 \\ 0 & 0 & -5 \\ 0 & -6 & -5 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 \\ 0 & 0 & 2 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & 1 & 1 \\ 3 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 1 \\ 0 & 2 & 0 \\ 6 & 2 & 2 \end{bmatrix}$$

### ii. Elementary matrix

One row operation away from identity

## b. Determinant

### i. Basics

#### 1) Property

$$\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$$

#### 2) Invertibility

If the corresponding vector transformation is invertible, the matrix product is invertible.

### ii. Method

$$\mathbf{A} \sim \mathbf{E}_{(-3)1,3} \mathbf{E}_{(-2)1,2} \begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 1 \\ 6 & -3 & 4 \end{bmatrix} \sim \mathbf{E}_{2,3} \begin{bmatrix} 2 & 1 & 3 \\ 0 & 0 & -5 \\ 0 & -6 & -5 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & 3 \\ 0 & -6 & -5 \\ 0 & 0 & -5 \end{bmatrix} = \mathbf{U}$$

$$\Rightarrow \mathbf{A} = \mathbf{E}_{(2)1,2} \mathbf{E}_{(3)1,3} \mathbf{E}_{3,2} \mathbf{U}$$

### iii. Inverse

We seldom calculate and store inverse.

We should store operation codes?

No, we store **LU**

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 3 & 1 & 1 \\ 3 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 1 \\ 0 & 2 & 0 \\ 6 & 2 & 2 \end{bmatrix}$$

# L6: **LU** Decomposition

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## 1. Inverse

### a. Decomposition of **LU**

#### i. Decomposition

$A = LU$ , where **L** is lower triangular and **U** is upper triangular

#### ii. Codes

Matlab: "[L,U]=lu(A)"

#### iii. Advantages of storing **LU** instead of $A^{-1}$

When **A** is sparse, **LU** is sparse as well, while  $A^{-1}$  is dense

### b. How to compute **LU**

#### i. Gauss' Elimination with condition

##### 1) Theorem 1: TYPE III

If all operations are TYPE III, take the **U** from gauss's elimination, corresponding  $\hat{L}$  will be lower triangle with unit main diagonal

##### 2) Theorem 2: Uniqueness

$$\left( \begin{array}{l} \det A \neq 0 \\ \exists_{\hat{L}, D, \hat{U}} A = \hat{L} D \hat{U} \end{array} \right) \Rightarrow (\hat{L}, D, \hat{U} \text{ are unique})$$

**D** is a diagonal matrix

##### 3) Theorem 3: Complementary of Theorem 1

$$\left( \begin{array}{l} \square \\ \det A \neq 0 \\ \square \end{array} \right) \Rightarrow \left( \begin{array}{l} \square \\ \forall_{k=1,2,\dots,n-1} \det A_k \neq 0 \Leftrightarrow \exists_{\hat{L}, U} A = \hat{L} U \\ \square \end{array} \right)$$

What if not only type III is carried out?

#### ii. Permutation matrix

##### 1) Definition, **P**

An identity matrix with reordered rows

##### 2) Property

$$P^{-1} = P^T$$

##### 3) Theorem 4: **P** is the concentration of row swap

$$\forall_{\det A \neq 0} \exists_{P, \hat{L}, U} A = P^T \hat{L} U$$

## 2. '

# L7: Structure & Iteration Method

2019年6月4日 12:15

1. '

- a. LU decomposition
- b. Special Structure
  - i. Tridiagonal system

1) Definition

$$\begin{bmatrix} f_1 & g_1 & & & \\ e_2 & f_2 & g_2 & & \\ & e_3 & f_3 & g_3 & \\ & & \ddots & \ddots & \ddots \\ & & & \ddots & \ddots & \ddots \\ & & & & e_{n-1} & f_{n-1} & g_{n-1} \\ & & & & & e_n & f_n \end{bmatrix}$$

2) Solution

Calculation is proportional to  $n$  instead of the  $n^3$  of general matrix

ii. Symmetric Positive Definite

1) Positive Definite

 a) Definition

$$\mathbf{A} \text{ is positive definite} \Leftrightarrow \forall_{\mathbf{x} \neq \mathbf{0}} f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} > 0$$

b) How to check whether it is positive definite

Check eigenvalue

2) Theorem

 a) Theorem

$$\forall \mathbf{A} \text{ is symmetric positive definite} \quad \exists \mathbf{C} \text{ such that } \mathbf{A} = \mathbf{C} \mathbf{C}^T$$

where  $\mathbf{C}$  is a lower triangular matrix with positive main diagonal

b) Usage for conceptual reason

We use theorem above to check whether a symmetric matrix is positive definite

c. Iteration Method for Solving  $\mathbf{Ax} = \mathbf{b}$

i. Jacobi Iteration

 1) Requirement

$\mathbf{D}$  is a diagonal matrix with  $\mathbf{A}$ 's diagonal element

$\mathbf{A}$  is invertible and has nonzero diagonal element

 2) Method

$$\mathbf{x}^{(k+1)} = \mathbf{D}^{-1}(\mathbf{b} + (\mathbf{D} - \mathbf{A})\mathbf{x}^{(k)})$$

3) Improvement: Updating Elements

- Instead of updating elements of  $\mathbf{x}^{(k)}$  in parallel,

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left( b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k)} - \sum_{j=i+1}^n a_{ij} x_j^{(k)} \right) \quad \text{for } i = 1, 2, \dots, n$$

we could update successively

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left( b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i+1}^n a_{ij} x_j^{(k)} \right) \quad \text{for } i = 1, 2, \dots, n$$

#### 4) Definition of Strictly Diagonally Dominant

$$\text{Strictly Diagonally Dominant} \Leftarrow \forall_{i=1,2,\dots,n} |a_{ii}| > \sum_{j=1}^{i-1} |a_{ij}| + \sum_{j=i+1}^n |a_{ij}|$$

#### 5) Theorem for convergence

$$\mathbf{A} \text{ is strictly diagonally dominant} \Rightarrow \left( \begin{array}{l} \mathbf{Ax} = \mathbf{b} \text{ exists a unique solution} \\ \forall_{\mathbf{x}^{(0)}} \{ \mathbf{x}^{(k)} \} \rightarrow \text{solution} \end{array} \right)$$

2. '

# L8: Orthogonality by Projection

2019年6月6日 12:14

## 1. Orthogonality by Projection

### a. Basics

#### i. Orthogonality

$\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$  are orthogonal  $\Leftrightarrow \mathbf{u}^T \mathbf{v} = 0$

#### ii. Span

A subspace of something

#### iii. How to find orthonormal basis for a space

Gram-Schmidt Process (We do projection)

#### iv. Notation

$\text{col}(\mathbf{A})$  is the span consist of column vectors of  $\mathbf{A}$

### b. Gram-Schmidt

#### i. Assumption

Input basis are linear independent

#### ii. By-Product

QR decomposition

#### ★iii. QR factorization

$$\mathbf{A}_{m \times n} = \mathbf{Q}_{m \times n} \mathbf{R}_{n \times n}$$

$\mathbf{Q}$  has orthonormal columns

$\mathbf{R}$  is an upper triangular matrix with positive diagonal entries

### c. Methods

#### i. Classic Method (CGS)

Reduce projection first, and do normalization at last

#### ii. Modified Method (MGS)

Do normalization each time after projection reduction

\*this alleviate errors caused by rounding

## L9: Householder

2019年6月11日 12:14

### 1. Orthogonality by Householder

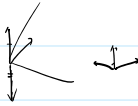
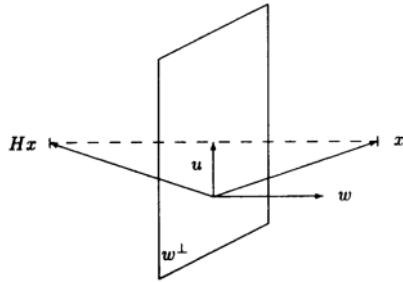
#### a. Tips

Orthogonal matrix is an transformation with length preserved

#### b. Material

<https://www.cnblogs.com/reasno/p/9606224.html>

#### c. Scheme



#### d. Equation

WHAT IS H

$$H_v = I_{n \times n} - \frac{2vv^T}{v^T v}$$

$$v = \begin{bmatrix} 0_{(j-1) \times 1} \\ \vdots \\ a_{(n-j+1) \times 1} \end{bmatrix} - ce_j \quad \text{where } c = \pm \|a_{(n-j+1) \times 1}\|$$

### 2. Eigenvalue

#### a. Introduction

Real practice do not do the  $\lambda$ :  $\det(A - I\lambda) = 0$  calculation

#### b. Similarity

##### i. Definition

$$\begin{pmatrix} \text{A, B are square matrix} \\ \Rightarrow \left( \text{B and A are similar} \right) \Leftrightarrow \left( \text{B} = \text{P}^{-1} \text{A} \text{P} \right) \end{pmatrix}$$

##### ★ ii. Property

Similar matrix have same eigenvalue

##### iii. Results

$\text{B} = \text{Q}^{-1} \text{A} \text{Q}$  and  $\text{A}$  have same eigenvalue

# Project

2019年6月12日 18:36

## 1. Material

### a. Root Finding

$$\mathbf{s}_k = \mathbf{x}_k - \mathbf{x}_{k-1}$$

$$\mathbf{y}_k = \mathbf{F}(\mathbf{x}_k) - \mathbf{F}(\mathbf{x}_{k-1})$$

$$\mathbf{A}_k^{-1} = \mathbf{A}_{k-1}^{-1} + \frac{(\mathbf{s}_k - \mathbf{A}_{k-1}^{-1} \mathbf{y}_k) \mathbf{s}_k^T \mathbf{A}_{k-1}^{-1}}{\mathbf{s}_k^T \mathbf{A}_{k-1}^{-1} \mathbf{y}_k}$$

$$\mathbf{w}_k = \mathbf{A}_k^{-1} \mathbf{F}(\mathbf{x}_k)$$

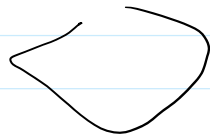
$$\mathbf{x}_{k+1} = \mathbf{x}_k - \mathbf{w}_k$$

$$\pi(x) = \exp\left(\sum_{j=0}^n \lambda_j x^j\right)$$

$$\underline{f(x)} = \int_0^1 \pi(x) dx = \sum_{i=0}^n \lambda_i m_i$$

f:

f(x)



$$\frac{\partial f(x)}{\partial \lambda_k} = \frac{\partial \int_0^1 \pi(x) dx}{\partial \lambda_k} = m_k$$

$$= \int_0^1 dx \left( \frac{\partial \exp\left(\sum_{j=0}^n \lambda_j x^j\right)}{\partial \lambda_k} \right) = m_k$$

$$= \int_0^1 dx \left( \frac{\partial}{\partial \lambda_k} \exp(\cdot) \cdot \exp(\lambda_k x^k) \right) = m_k$$

$$= \int_0^1 \underbrace{x^k \pi(x)}_{\lambda_k} dx = \underbrace{m_k}_{\lambda_k}$$

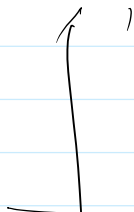
$$f = g: \mathbb{R}^n \rightarrow \mathbb{R}$$

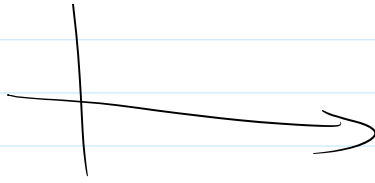
$$g(x_1, x_2, \dots, x_n) = y$$

$\bigwedge_{j=1,2,\dots,n}$ 
 $\bigwedge_{a \leq b}$

$$\left. \frac{\partial g}{\partial x_i}(x_1, x_2, \dots, x_n) \right|_{x_i=a} = \frac{\partial g}{\partial x_i}(x_1, x_2, \dots, b, \dots, x_n)$$

1b





$$f(\lambda) = \int_0^1 \exp\left(\sum_{j=0}^n \lambda_j x^j\right) dx - \sum_{i=0}^n \lambda_i m_i = \int_0^1 \pi(x) dx - \sum_{i=0}^n \lambda_i m_i$$

$$\nabla f(\lambda) = \begin{bmatrix} \frac{\partial f(\lambda)}{\partial \lambda_0} \\ \frac{\partial f(\lambda)}{\partial \lambda_1} \\ \vdots \\ \frac{\partial f(\lambda)}{\partial \lambda_n} \end{bmatrix} = \begin{bmatrix} \int_0^1 x^0 \pi(x) dx - m_0 \\ \int_0^1 x^1 \pi(x) dx - m_1 \\ \vdots \\ \int_0^1 x^n \pi(x) dx - m_n \end{bmatrix}$$

We need to find  $F(\lambda) \stackrel{\text{def}}{=} \nabla f(\lambda) = \vec{0}$

$$\frac{\partial f(\lambda)}{\partial \lambda_i} = \frac{\partial}{\partial \lambda_i} \left( \int_0^1 \pi(x, \lambda) dx \right) - \frac{\partial}{\partial \lambda_i} \left( \sum_{i=0}^n \lambda_i m_i \right)$$

$$\frac{\partial}{\partial \lambda_i} \left( \int_0^1 \pi(x, \lambda) dx \right) = \int_0^1 \frac{\partial \pi(x, \lambda)}{\partial \lambda_i} dx$$

# Presentation

2019年8月2日 20:47

## 1. Requirement

Speak clear for 3 minutes

## 2. Outline: Why we choose Newton's method for Task 1

a. Very likely to be convex

b. Not much operation

## 3. Speech

|   |                                                                                                                                                                                            |  |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 1 | In next 3 minutes, I will explain why and how we apply in task 1 Newton's method. i.e. does the mathematical precondition for newton's method hold? And is numerical operation affordable. |  |
|   | Let's write down the hessian, using this notation to represent partial derivative and mixed derivative                                                                                     |  |
|   | Through analytical calculation, we find some nice property of hessian.                                                                                                                     |  |
|   | First, Hessian is a Hankel Matrix: all elements in this direction share same value.                                                                                                        |  |
|   | Second, all elements are strictly position and strictly increasing in this direction                                                                                                       |  |
| 2 | Based on those property, we spent effort proving convexity of $f(\lambda)$ by proving postive definiteness of <b>H</b> .                                                                   |  |
|   | One reference gave this sufficient condition for positive definiteness of Hankel matrix:                                                                                                   |  |
|   | If there exists such a measure $\sigma$ where all elements                                                                                                                                 |  |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <p><math>h_i</math> in Hankel can be represented in such ways, then the Hankel has positive definiteness.</p> <p>We think this theorem lead us to the door of proving positive definiteness in our case. But we are still not strong enough to prove whether there exists such a measure. So we take engineering approach. We try Newton's method.</p>                                                                                                                                                                                                                                                                                                          |
| 3 | <p>For engineer, another question is also very important: Are numerical operations affordable in Newton's method.</p> <p>We notice that, in this question, the main complexity is in evaluating <math>h_i</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4 | <p>Taking advantage of nice property of <math>h_i</math>, we find a cyclic ways for such evaluation. We take <math>p</math> as the step of Riemann Sum.</p> <p>We construct <math>TEMP_i</math> vector such that any <math>h_i</math> is the sum of elements of <math>TEMP_i</math></p> <p>Then, <math>TEMP_0</math> is simply doing this operation to <math>x_p</math> pointwisely.</p> <p>Notice that <math>h_{i+1}</math> is simply multiplying one more <math>x</math> in integral, thus <math>TEMP_{i+1}</math> equals <math>TEMP_i</math> multiplied pointwisely by <math>x_p</math></p> <p>In this way, we reduce complexity by reusing calculation.</p> |
| 5 | <p>As a result, computing <math>f(\lambda)</math> requires that much operation, while computing gradient and hessian only requires 3 times that much</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|   |                                                               |  |
|---|---------------------------------------------------------------|--|
|   |                                                               |  |
| 6 | Solving Hankel linear system takes another $O(n^2)$ operation |  |

# L10: Euler's Method

2019年6月13日 12:16

## 1. Ordinary Differential Equation

### a. Question

$$\dot{y} = \Phi(t, y), y(t_0) = y_0, t \in [t_0, T]:$$

solve for  $y(t)$

### b. Well-Posed $\Phi$

$\phi$  is continuous and satisfies Lipschitz condition on  $\mathcal{D} = \{(t, y) : t \in [t_0, T], y \in (-\infty, +\infty)\}$

### c. Euler's Method

#### i. The method

We use  $\hat{y}_k$  to approximate  $y(t_k)$ , where

$$\begin{cases} t_k = t_0 + kh, & k \in \mathbb{N} \\ \hat{y}_0 = y_0 \\ \hat{y}_k := \text{eular}(t_{k-1}, \hat{y}_{k-1}) = \hat{y}_{k-1} + h\Phi(t_{k-1}, \hat{y}_{k-1}) \end{cases}$$

#### ii. Error

##### 1) Local Discretization Error

$$\tau_k = y(t_k) - \widetilde{y}_k := y(t_k) - \text{eular}(t_{k-1}, y_{k-1})$$

Error caused by local linear interpretation

##### 2) Global Discretization Error

$$e_k = \hat{y}_k - y(t_k)$$

Error accumulated by global iteration

##### 3) Theorem: bounded global error under well-posed

$$\left( \begin{array}{c} \square \\ \text{well posed } \phi(t, y) \\ \square \end{array} \right) \Rightarrow \left( \begin{array}{c} \square \\ |e_k| \leq \frac{\tau^*}{hc} (e^{c(t_k - t_0)} - 1) \\ \tau^* = \max_k |\tau_k|, \text{ } c \text{ is the lipschitz constant} \end{array} \right)$$

#### iii. Problem

##### 1) Stability

##### 2) Convergence

## 2. '

# L11: Development of Euler

2019年6月18日 12:17

1. '

a. Backward Euler's Method

i. Method

$$y_{k-1} = y_k - h\Phi(t_k)$$

ii. Advantage

Avoid overshooting

High stability, but also high cost for solving equation

b. Generalization of Euler's Method

i. Multiderivative Method

1) Method

Second order Taylor's expansion

ii. '

2. '

# L12: Farewell to Jing

2019年6月20日 12:17

## 1. ODE

### a. RK-4

#### i. Why we need multistage

To approximate Taylors' method without explicitly evaluating derivatives

#### ii. Simpson's Rule

在此处键入公式。

A more sophisticated way of estimating integral

### b. BVP

#### i. Difference from IVP

Conditions are spread on boundary instead of within one point

#### ii. ;

## 2. Exam

### a. Short Answer Question: Understanding Question

#### i. T&F

#### ii.

### b. Proof Question

#### i. 1

## 1. Root Finding

## a. Basics

$$\left( \underbrace{\left( \exists_{c \in \mathbb{R}} \forall_{x_1, x_2} |f(x_1) - f(x_2)| \leq c|x_1 - x_2| \right)}_{\substack{\text{Continuously} \\ \text{Differentiable} \subseteq \text{Differentiable} \not\subseteq \text{Lipschitz Continuous} \subseteq \text{Continuous}}} \right) \xrightarrow{\text{define}} \text{Lipschitz continuous}$$

## b. Bisection Method

## i. Requirement:

Continuity &amp; Proper initial interval

## ii. Theorem behind: Intermediate Value Theorem

$$(f: [a, b] \rightarrow \mathbb{R} \text{ is continuous}) \Rightarrow \left( \forall_{k \in (f(a), f(b))} \exists_{x \in (a, b)} f(x) = k \right)$$

c. Fixed Point Iteration:  $x_{k+1} = g(x_k)$ 

## i. Requirement: Lipschitz Continuity

 $f(x) = g(x) - x$ , where  $g(x)$  has Lipschitz constant  $c \in (0, 1)$ 

## ii. Condition for quadratic convergence

 $g(x)$  need to be twice differentiabled. Newton's Method:  $x_{k+1} := x_k - \frac{f(x_k)}{f'(x_k)}$ 

## i. Theorem

$$\left( f \in C^2[a, b] \quad \exists_{x^* \in [a, b]} f(x^*) = 0 \right) \Rightarrow \left( f'(x^*) \neq 0 \Rightarrow \left( \exists_{\delta > 0} \forall_{x_0 \in [x^* - \delta, x^* + \delta]} \left\{ x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \right\}_{n=0}^{\infty} \xrightarrow{\text{quadratically}} x^* \right) \right)$$

ii. Initial Position:  $\left| \frac{f(x) \cdot f''(x)}{f'(x)^2} \right| < 1$ 

## e. Secant Method

Substitute  $\frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}$  for  $f'(x_k)$  in Newton's Method

## 2. Optimization

## a. Basics

## i. First Order: Fermat Theorem:

$$\left( \begin{array}{c} \square \\ \text{once differentiable} \\ \square \end{array} \right) \Rightarrow \left( \begin{array}{c} \square \\ [\text{local minimum at } x^*] \Rightarrow [f'(x^*) = 0] \\ \square \end{array} \right)$$

## ii. Theorem: Sufficient and Necessary first order condition

$$\left( \begin{array}{c} f: \mathbb{R} \rightarrow \mathbb{R} \\ \text{convex} \\ \text{differentiable} \end{array} \right) \Rightarrow \left( \begin{array}{c} \square \\ \text{global minimum at } x^* \Leftrightarrow f'(x^*) = 0 \\ \square \end{array} \right)$$

## iii. Second Order:

$$\left( \begin{array}{c} \text{twice continuously} \\ \text{differentiable} \\ \text{in open neighbourhood} \end{array} \right) \Rightarrow \left( \begin{array}{c} \left\{ \begin{array}{l} f'(x^*) = 0 \\ f''(x^*) > 0 \end{array} \right\} \Rightarrow [\text{local minimum at } x^*] \end{array} \right)$$

iv. Convexity

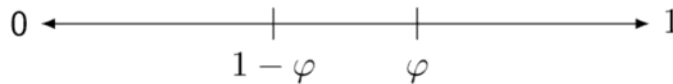
$$\forall x_1, x_2 \in \mathbb{R}^n \quad \forall \alpha \in [0, 1] \quad f(\alpha x_1 + (1 - \alpha)x_2) \leq \alpha f(x_1) + (1 - \alpha)f(x_2)$$

v. Unimodal

$$\exists x^* \in J \quad \text{strictly decreasing for } [a, x^*], \text{ strictly increasing for } [x^*, b]$$

b. Newton:  $x_p = x_k - \frac{f'(x_k)}{f''(x_k)}$

c. Golden Section Search:  $\varphi = \frac{-1 + \sqrt{5}}{2}$



### 3. Linear System

a. Basics

|            |                                         |                   |                                         |
|------------|-----------------------------------------|-------------------|-----------------------------------------|
| Type I :   | Interchange two rows                    | $E_{i,j}$         | $\det(E_{i,j}A) = -\det(A)$             |
| Type II :  | Multiply a row by a nonzero constant    | $E_{(\alpha)i}$   | $\det(E_{(\alpha)i}A) = \alpha \det(A)$ |
| Type III : | Add a constant times one row to another | $E_{(\alpha)i,j}$ | $\det(E_{(\alpha)i,j}A) = \det(A)$      |

$$\det(AB) = \det A \det B$$

b. LU decomposition

i. Theorem 1: TYPE III

If all operations are TYPE III, take the  $U$  from gauss's elimination, corresponding  $\hat{L}$  will be lower triangle with unit main diagonal

ii. Theorem 2: Uniqueness

$$\left( \begin{array}{c} \det A \neq 0 \\ \exists \hat{L}, \hat{D}, \hat{U} \end{array} \right) \Rightarrow (\hat{L}, \hat{D}, \hat{U} \text{ are unique}), \hat{D} \text{ is a diagonal matrix}$$

iii. Theorem 3: Complementary of Theorem 1

$$\left( \begin{array}{c} \det A \neq 0 \end{array} \right) \Rightarrow \left( \begin{array}{c} \forall_{k=1,2,\dots,n-1} \det A_k \neq 0 \Leftrightarrow \exists \hat{L}, \hat{U} \end{array} \right)$$

iv. Theorem 4:  $P$  is the concentration of row swap

$$\forall_{\det A \neq 0} \exists_{P, \hat{L}, \hat{U}} A = P^T \hat{L} \hat{U}$$

c. Symmetric Positive Definite (Then we can do Cholesky)

$$A = CC^T, \quad C \text{ is a lower triangular matrix with positive main diagonal}$$

d. Iteration Method for Solving  $Ax = b$

i. Requirement

$A$  is invertible and has nonzero diagonal element

ii. Method

$$x^{(k+1)} = D^{-1}(b + (D - A)x^{(k)})$$

iii. Requirement for uniqueness, existence and convergence

$A$  is strictly diagonally dominant

### 4. Orthogonality

a. By Projection

$$\mathbf{A}_{m \times n} = \mathbf{Q}_{m \times n} \mathbf{R}_{n \times n}$$

$\mathbf{Q}$  has orthonormal columns

$\mathbf{R}$  is an upper triangular matrix with positive diagonal entries

b. By Reflection

? Householder

c. Similarity

$$(\mathbf{B} \text{ and } \mathbf{A} \text{ share same eigenvalue}) \Leftrightarrow (\mathbf{B} \text{ and } \mathbf{A} \text{ are similar}) \Leftrightarrow (\mathbf{B} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P})$$

## 5. Integration

a. Well-Posed  $\Phi$

$\phi$  is continuous and satisfies Lipschitz condition on  $\mathcal{D} = \{(t, y) : t \in [t_0, T], y \in (-\infty, +\infty)\}$

b. Euler's Method:

i. Equation:

$$\hat{y}_k := \text{euler}(t_{k-1}, \hat{y}_{k-1}) = \hat{y}_{k-1} + h\Phi(t_{k-1}, \hat{y}_{k-1})$$

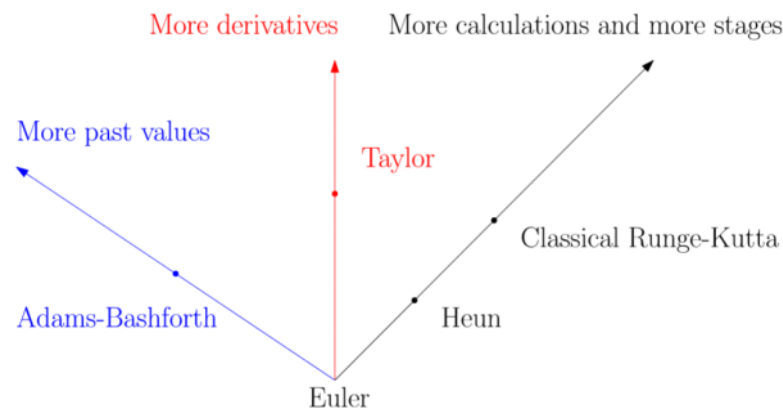
ii. Error:

|                  |                                       |
|------------------|---------------------------------------|
| Local $\tau_k$ : | Caused by local linear interpretation |
| Global $e_k$ :   | Accumulated by global iteration       |

iii. Theorem

$$\left( \begin{array}{c} \square \\ \text{well posed } \phi(t, y) \\ \square \end{array} \right) \Rightarrow \left( \begin{array}{c} \square \\ |e_k| \leq \frac{\tau^*}{hc} (e^{c(t_k - t_0)} - 1) \\ \tau^* = \max_k |\tau_k|, \text{ } c \text{ is the lipschitz constant} \end{array} \right)$$

c. Generalization



# L13: Preview of Structure

2019年6月25日 12:17

## 1. Preview

### a. Basic

#### i. Group, $(G, \circ)$ , $\circ : G \times G \rightarrow G$

##### 1) Definition

A pair of set and one operator with **associativity**, **unit 0**, **inverse**

##### 2) Abelian

If **abelian**, have **commutativity**

##### 3) Attention (Optional)

May not have distributivity

#### ii. Ring, $(R, +, \cdot)$ , $+, \cdot : R \times R \rightarrow R$

##### 1) Definition

a)  $(R, +)$  is **abelian group**

b)  $\cdot$  has **unit 1**, **associativity** and **distributivity**

##### 2) Commutative

If **commutative**, then have commutativity

#### iii. Field, $(F, +, \cdot)$

##### 1) Definition

a)  $(F, +, \cdot)$  is a **commutative ring**

b)  $0 \neq 1$  and for nonzero elements have inverse

##### 2) Attention

#### iv. Vector Space $(V, +, \cdot)$ over Field $F$

##### 1) Definition of Vector Space

### Definition (Vector space)

A *vector space*  $(V, +, \cdot)$  over a field  $F$  is a set on which the  $+$  and  $\cdot$  operations are defined and closed. Moreover, for any  $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$  and  $a, b \in F$  the following conditions hold:

i *Commutativity for the addition:*  $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ ;

ii *Associativity for the addition:*  $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$ ;

iii *Additive identity:* there exists  $\mathbf{0} \in V$  such for any  $\mathbf{v} \in V$ ,  $\mathbf{0} + \mathbf{v} = \mathbf{v} + \mathbf{0} = \mathbf{v}$ ;

iv *Additive inverse:* for any  $\mathbf{v} \in V$ , there exists  $-\mathbf{v} \in V$  such that  $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$ ;

v *Commutativity of scalar multiplication:*  $a \cdot (b\mathbf{v}) = (ab) \cdot \mathbf{v}$ ;

vi *Distributivity of scalar multiplication for vector addition:*

$$a \cdot (\mathbf{u} + \mathbf{v}) = a \cdot \mathbf{u} + a \cdot \mathbf{v};$$

vii *Distributivity of scalar multiplication for field addition:*

$$(a + b) \cdot \mathbf{v} = a \cdot \mathbf{v} + b \cdot \mathbf{v};$$

viii *Scalar multiplication identity:* for all  $\mathbf{v} \in V$ ,  $1_F \cdot \mathbf{v} = \mathbf{v}$ ;

## 2) Terminology

### a) Basis

Basis of  $V$  is a subset of  $V$

b) Finite Dimension

If  $V$  is spanned by finite number of vectors

c) Linear Map / Linear Application  $f: V_1 \rightarrow V_2$

$$\forall_{x,y \in V_1, a \in F}$$

$$f(x + y) = f(x) + f(y)$$

$$f(ax) = af(x)$$

d) Kernel of  $f$ ,  $\ker f$

$$f: E \rightarrow F$$

$$\ker f = \{x: f(x) = 0, x \in E\}$$

3) Lemma: Bijective and Kernel

For vector spaces  $E$  and  $F$  with same dimension, linear mapping  $f: E \rightarrow F$  is bijective  $\Leftrightarrow$

$$\ker f = \{0\}$$

# L14: Vector Algebra

2019年6月27日 12:13

## 1. Vector Space

### a. Terminology

#### i. Basis

Basis of  $V$  is a subset of  $V$

#### ii. Finite Dimension

If  $V$  is spanned by finite number of vectors

#### iii. Linear Map / Linear Application $f: V_1 \rightarrow V_2$

$$\forall x, y \in V_1, a \in F$$

$$f(x + y) = f(x) + f(y)$$

$$f(ax) = af(x)$$

#### iv. Kernel of $f$ , $\ker f$

$$f: E \rightarrow F$$

$$\ker f = \{x: f(x) = 0, x \in E\}$$

### b. Lemma: Bijective and Kernel

For vector spaces  $E$  and  $F$  with same dimension, linear mapping  $f: E \rightarrow F$  is bijective  $\Leftrightarrow$

$$\ker f = \{0\}$$

### c. Inner Product and Norm

#### i. Definition

#### Definitions

Let  $V$  be a vector space over a field  $\mathbb{K}$ .

① The map  $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{K}$  is called an *inner product* if for  $u, v, w \in V$  and  $a \in \mathbb{K}$  it satisfies:

① *Conjugate symmetry*:  $\langle u, v \rangle = \overline{\langle v, u \rangle}$ ;

② *Linearity*:  $\langle au, v \rangle = a\langle u, v \rangle$  and  $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$ ;

③ *Positivity*:  $\langle u, u \rangle \geq 0$  and  $\langle u, u \rangle = 0 \Leftrightarrow u = 0$ ;

A vector space with an inner product is an *inner product space*.

② The map  $\| \cdot \| : V \rightarrow \mathbb{R}_+$  is called a *norm* if for any  $u, v \in V$  and  $a \in \mathbb{K}$  it satisfies:

①  $\|u\| = 0 \Leftrightarrow u = 0$ ;

②  $\|au\| = |a|\|u\|$ ;

③ *Triangular inequality*:  $\|u + v\| \leq \|u\| + \|v\|$ ;

A vector space with a norm is an *normed vector space*.

#### ii. Inner Product $\Rightarrow$ Norm

$$\|u\| := \sqrt{\langle u, u \rangle}$$

#### iii. Theorem: Cauchy-Schwarz Inequality

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$$

#### iv. Parallelogram Law

##### Proposition

Let  $E$  be an inner-product space over  $\mathbb{R}$  and  $u, v \in E$ .

- ① Parallelogram law:  $\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2)$ ;
- ②  $\|u + v\|^2 - \|u - v\|^2 = 4\langle uv \rangle$ ;

#### v. Norm $\Rightarrow$ Inner Product

##### Remark.

Most normed vector spaces over  $\mathbb{R}$  and  $\mathbb{C}$  do not have an inner-product. In other words, the converse of proposition 4.21 is false. However if for a norm the parallelogram law is verified then this norm arises from an inner-product.

#### d. Norm of Map

##### i. Definition

- 1) Requirement:  $u: E \rightarrow F$

$u$  is a linear map between two normed space in  $\mathbb{K}$

- 2) Equation

$$\|u\| := N(u) := \sup_{x \in E \setminus \{0\}} \frac{\|u(x)\|}{\|x\|}$$

##### ii. Set of Norm

$$L(E, F) := \{u\}$$

$$\mathcal{L}(E, F) := \{u: \|u\| < \infty\}$$

##### iii. Theorem: Inherence of Convergence

##### Proposition

Let  $E$  and  $F$  be two vector spaces over  $\mathbb{K}$  and  $u \in \mathcal{L}(E, F)$ . If  $(x_n)_{n \in \mathbb{N}}$  is a sequence converging to  $l$ , then the sequence  $(u(x_n))_n$  converges to  $u(l)$ .

#### e. Metric Space and Distance

##### i. Definition

- ? Does norm naturally define a distance?
- ? Does distance naturally define a norm?

## Definitions

- ① Let  $X$  be a set. A *distance* on  $X$  is a function  $d : X \times X \rightarrow \mathbb{R}_+$  such that for all  $x, y, z \in X$ :
  - i  $d(x, y) = 0$  if and only if  $x = y$ ;
  - ii  $d(x, y) = d(y, x)$ ;
  - iii  $d(x, z) \leq d(x, y) + d(y, z)$  (triangular inequality);
 A set endowed with a distance is called a *metric space*.
- ② Let  $X$  and  $Y$  be two metric spaces. A function  $f : X \rightarrow Y$  is *continuous* in  $a$ , if for all  $\varepsilon \in \mathbb{R}_+^*$ , there exists  $\eta \in \mathbb{R}_+^*$ , such that  $d(a, x) < \eta$  implies  $d(f(a), f(x)) < \varepsilon$ .
- ③ Let  $X$  and  $Y$  be two metric spaces. A function  $f : X \rightarrow Y$  is *continuous* if it is continuous on any point of  $X$ .

## ii. Continuity on $\mathbb{R}$

*Lipschitz*  $\subset$  *absolutely*  $\subset$  *uniformly*  $\subset$  *continuous*

? Why is absolutely continuous more demanding than uniformly continuous?

## Definitions

Let  $X$  and  $Y$  be two metric spaces, and  $f : X \rightarrow Y$ .

- ① The function  $f$  is *uniformly continuous* if for all  $\varepsilon \in \mathbb{R}_+^*$ , there exists  $\eta \in \mathbb{R}_+^*$ , such that  $d(x, y) < \eta$  implies  $d(f(x), f(y)) < \varepsilon$ , for  $x, y \in X$ .
- ② The function  $f$  is *Lipschitz continuous* if there exists  $k$  such that for any  $x, y \in X$ ,  $d(f(x), f(y)) \leq kd(x, y)$ .
- ③ Let  $X$  be an interval of  $\mathbb{R}$ . The function  $f$  is *absolutely continuous* if for all  $\varepsilon \in \mathbb{R}_+^*$ , there exists  $\eta \in \mathbb{R}_+^*$ , such that whenever  $\{[x_i, y_i] : i = 1, \dots, n\}$  is a finite collection of mutually disjoint subintervals of  $X$ ,  $\sum_{i=1}^n |y_i - x_i| < \eta$  implies  $\sum_{i=1}^n d(f(x_i), f(y_i)) < \varepsilon$ .

# L15: Differential & Topology

2019年7月2日 12:17

## 1. Differentiability

### a. Basics

#### i. Theorem

##### Theorem

Let  $E$  and  $F$  be two normed vector spaces over  $\mathbb{K}$  and  $u$  be a linear application. The following conditions are equivalent.

- i  $u \in \mathcal{L}(E, F)$ ;
- ii  $u$  is uniformly continuous;
- iii  $u$  is continuous;
- iv  $u$  is continuous for any  $x \in E$ ;
- v  $u$  is bounded over any subset of  $E$ ;
- vi There exists  $a \in E$  and  $r \in \mathbb{R}_+^*$  such that  $u$  is bounded over  $B(a, r)$ ;

Proof of vi  $\rightarrow$  i tries to use local property to prove uniform property. This requires us to map the ball  $B(a, r)$  to  $\mathbb{R}$

#### ii. Null Function Lemma

$$u \in L(E, F) \Rightarrow \left( \lim_{x \rightarrow 0} \frac{u(x)}{\|x\|} \Rightarrow u(x) = 0 \right)$$

### b. Differential

#### i. Theorem: Existence and Uniqueness

There exists at most one linear application  $\ell \in$

$$L(E, F): \lim_{x \rightarrow 0} \frac{f(a+x) - f(a) - \ell(x)}{\|x\|} = 0$$

Why is it  $\ell(x)$ ? Why is it not  $\ell(a)$ ?

#### ii. Definition of Differential

### Definitions

- ① When it exists, the linear application  $\ell$  is called the *differential of  $f$  in  $a$* . Then the function  $f$  is said to be *differentiable in  $a$* . The differential of  $f$  in  $a$  is denoted  $df(a)$ .
- ② If for all  $a \in U \subset E$ ,  $df(a)$  exists then we define the application

$$\begin{aligned} df : U &\longrightarrow L(E, F) \\ x &\longmapsto df(x). \end{aligned}$$

This application is called the *differential of  $f$* .

- ③ When  $f$  is  $n$  times differentiable on  $U \subset E$ ,  $n \in \mathbb{N}^*$ , the differential of order  $n$  of  $f$  is  $n$ -linear and given by

$$\begin{aligned} d^n f : U &\longrightarrow L_n(E, F) = L(E, L_{n-1}(E, F)) \\ x &\longmapsto d^n f(x). \end{aligned}$$

iii. Tips

Differential is a linear map between maps

## 2. Topology

### a. Definition

#### i. Definition of Topological Space $(X, \mathcal{T})$

### Definitions

- ① A topological space is a pair  $(X, \mathcal{T})$  where  $X$  is a set and  $\mathcal{T}$  is a family of subsets of  $X$  called the *topology* of  $X$ . Its elements are called *open sets* and satisfy the following three properties:
  - i  $\emptyset, X \in \mathcal{T}$ ;
  - ii Any union, finite or not, of open subsets of  $X$  is open;
  - iii The intersection of finitely many open subsets of  $X$  is open;
- ② Let  $x$  be in  $X$ . Any open set containing  $x$  is called a *neighborhood* of  $x$ .

- 1) A topology space is a pair  $(X, \mathcal{T})$ , where  $X$  is a set, and  $\mathcal{T}$  is a family of subset of  $X$ .
- 2) Elements of  $\mathcal{T}$  are called open sets and satisfied:
  - a)  $\emptyset, X \in \mathcal{T}$
  - b) Any union of open subsets of  $X$  is open
  - c) Finite intersection of open subset of  $X$  is open

## ii. Definition of Neighborhood

Let  $x$  be in  $X$ . Any open set containing  $x$  is called a neighborhood of  $x$

## iii. Example

- If all the subsets of  $\mathbb{Z}$  are viewed as open then  $\mathbb{Z}$  becomes a topological space.
- If the open sets of  $\mathbb{R}$  are defined in the usual way then  $\mathbb{R}$  is a topological space.
- If we decide that the open sets of  $\mathbb{R}$  are  $\emptyset$ ,  $\mathbb{R}$ , and all the half lines  $\{x \in \mathbb{R} : x \geq a \text{ and } a \in \mathbb{R}\}$ , then  $\mathbb{R}$  is not a topological space.

## b. Metric Space

### i. Theorem:

A metric space is a topology space

### ii. Definition

#### Definitions

- ① Let  $X$  be a metric space. A subset of  $X$  is said to be *open* if it can be written as a union of open balls.
- ② Let  $X$  be a metric space. A subset  $U$  of  $X$  is said to be *closed* if  $X \setminus U$  is an open subset of  $X$ .

### iii. Theorem

Only continuous mapping return the close/open property from "TO" to "FROM"

#### Theorem

Let  $X$  and  $Y$  be two metric spaces and  $f : X \rightarrow Y$ . Then the following conditions are equivalent

- i  $f$  is continuous;
- ii For any open  $V \subset Y$ ,  $f^{-1}(V)$  is an open of  $X$ ;
- iii For any closed  $V \subset Y$ ,  $f^{-1}(V)$  is closed in  $X$ ;

## c. Connected Space

### i. Definition

#### Definition

Let  $X$  be a metric space. If the only subsets of  $X$  to be both open and closed are  $X$  and  $\emptyset$ , then  $X$  is *connected*.

### ii. Intermediate Value Theorem

Map from connected metric space obey Intermediate Value Theorem from "FROM" to "TO"

**Theorem** (Intermediate value theorem)

Let  $X$  be a connected metric space, and  $a, b \in X$ . If  $f : X \rightarrow \mathbb{R}$  is continuous then it takes all the values between  $f(a)$  and  $f(b)$ .

d. Compact Space

i. Definition

A compact space can be covered by finite subset

**Definitions**

- ① A collection  $U$  is called an *open cover* of a topological space  $X$  if the elements of  $U$  are open subsets of  $X$  and the union of all elements in  $U$  is  $X$ .
- ② A space  $X$  is said to be *compact* if every open cover of  $X$  contains a finite subcover.

ii. Theorem

Continuous mapping send compact property from  
"FROM" to "TO"

**Theorem**

Let  $X$  and  $Y$  be two metric spaces and  $f : X \rightarrow Y$  be a continuous function. Then for any compact subset  $A \subset X$ ,  $f(A) \subset Y$  is compact.

# L16: Connected, Compact & Convergence

2019年7月9日 12:12

## 1. Topology

### a. Connected Space

#### i. Definition

##### Definition

Let  $X$  be a metric space. If the only subsets of  $X$  to be both open and closed are  $X$  and  $\emptyset$ , then  $X$  is *connected*.

#### ★ ii. Intermediate Value Theorem

Require Connected Space

##### Theorem (Intermediate value theorem)

Let  $X$  be a *connected* metric space, and  $a, b \in X$ . If  $f : X \rightarrow \mathbb{R}$  is continuous then it takes all the values between  $f(a)$  and  $f(b)$ .

### b. Compact

#### i. Definition

Compact means by no means can one find a infinite cover

##### Definitions

- ① A collection  $U$  is called an *open cover* of a topological space  $X$  if the elements of  $U$  are open subsets of  $X$  and the union of all elements in  $U$  is  $X$ .
- ② A space  $X$  is said to be *compact* if every open cover of  $X$  contains a finite subcover.

#### ii. Property

An open interval cannot be compact

#### iii. Theorem on continuity

Continuous map send "Compact" from "FROM" to "TO"

##### Theorem

Let  $X$  and  $Y$  be two metric spaces and  $f : X \rightarrow Y$  be a continuous function. Then for any compact subset  $A \subset X$ ,  $f(A) \subset Y$  is compact.

#### ★ iv. Extreme Value Theorem

### Theorem (Extreme value)

Let  $X$  be a metric space,  $f : X \rightarrow \mathbb{R}$  a continuous map, and  $A$  be a non-empty compact of  $X$ . Then  $f$  is bounded and reaches both its upper and lower bounds.

#### v. Corollary

### Corollary

If  $A$  be a compact subset of a metric space  $X$ , then  $A$  is bounded.

Because distance function is continuous

#### vi. Bolzano-Weierstrass

Converge to an element in  $A$

### Theorem (Bolzano-Weierstrass)

A subset  $A$  of  $\mathbb{R}^n$ ,  $n \in \mathbb{N}$ , is compact if and only if any sequence in  $A$  has a sub-sequence converging to an element in  $A$ .

#### ★ vii. Heine-Borel Theorem

In real vector space, closed and bounded interval = compact

### Theorem (Heine-Borel)

A subset of  $\mathbb{R}^n$ ,  $n \in \mathbb{N}$ , is compact if and only if  $A$  is closed and bounded.

#### viii. Rolle's Theorem

### Theorem (Rolle's theorem)

Let  $f$  be a  $C^n[a, b]$  function, for  $a, b \in \mathbb{R}$ , which has  $n + 1$  distinct roots in  $[a, b]$ . Then there exists  $c \in (a, b)$  such that  $f^{(n)}(c) = 0$ .

#### c. About Convergence

##### i. Definition of Banach Space

Is there a space where Cauchy sequence does not converge?  
Some distance function does not satisfy completeness

### Definitions

Let  $X$  be a metric space.

- ① A sequence  $(x_n)_n$  is a *Cauchy sequence* if for any  $\varepsilon > 0$ , there exists  $N \in \mathbb{N}$  such that  $d(x_m, x_k) \leq \varepsilon$  when  $m, k \geq N$ .
- ② If any Cauchy sequence on  $X$  converges, then  $X$  is *complete*.
- ③ A *Banach space* is a complete normed vector space.

### ★ ii. Banach-Steinhaus Theorem (Very strong, iff)

#### Theorem (Banach-Steinhaus)

Let  $E$  be a Banach space,  $F$  be a normed vector space over  $\mathbb{K}$ , and  $H$  be a non-empty subset of  $\mathcal{L}(E, F)$ . Then  $\sup\{\|u\| : u \in H\}$  is finite if and only if for all  $x \in E$ ,  $\sup\{\|u(x)\| : u \in H\}$  is also finite.

### iii. Definition of Subalgebra

#### Definition of uniform convergence

#### Definitions

- ① Let  $\mathcal{E}$  be a set and  $\mathcal{F}(\mathcal{E}, \mathbb{K})$  be the set of the functions defined from  $\mathcal{E}$  into  $\mathbb{K}$ .  
The set  $\mathcal{H}$  is a *subalgebra* of  $\mathcal{F}(\mathcal{E}, \mathbb{K})$  if
  - i  $\mathcal{H}$  is a vector-subspace of  $\mathcal{F}(\mathcal{E}, \mathbb{K})$ ;
  - ii The constant applications are in  $\mathcal{H}$ ;
  - iii If  $f, g \in \mathcal{H}$ , then for all  $x$ ,  $f(x)g(x)$  is also in  $\mathcal{H}$ ;
- ② The subalgebra  $\mathcal{H}$  *separates*  $\mathcal{F}(\mathcal{E}, \mathbb{K})$ , if for any two distinct elements  $x, y \in \mathcal{E}$ , there exists  $u \in \mathcal{H}$  such that  $u(x) \neq u(y)$ .
- ③ Let  $X$  be a metric space. A sequence of function  $(f_n)_n$ , *uniformly converges*, if it converges in  $\mathcal{F}(\mathcal{E}, X)$ .

### iv. Stone-Weierstrass Theorem

#### ★ 1) Theorem

#### Theorem (Stone-Weierstrass)

Let  $X$  be a compact metric space and  $\mathcal{H}$  be a subalgebra that separates  $C(X, \mathbb{R})$ . Then any element from  $C(X, \mathbb{R})$  is the uniform limit of a sequence of elements from  $\mathcal{H}$ .

#### 2) Corollary

#### Base of Taylor Theorem

### Corollary

Any continuous function defined over  $[a, b] \subset \mathbb{R}$  is the uniform limit of a sequence of polynomials.

# L17: Measure

2019年7月11日 10:53

## 1. Integral

### a. Riemann Sum

#### i. Definition

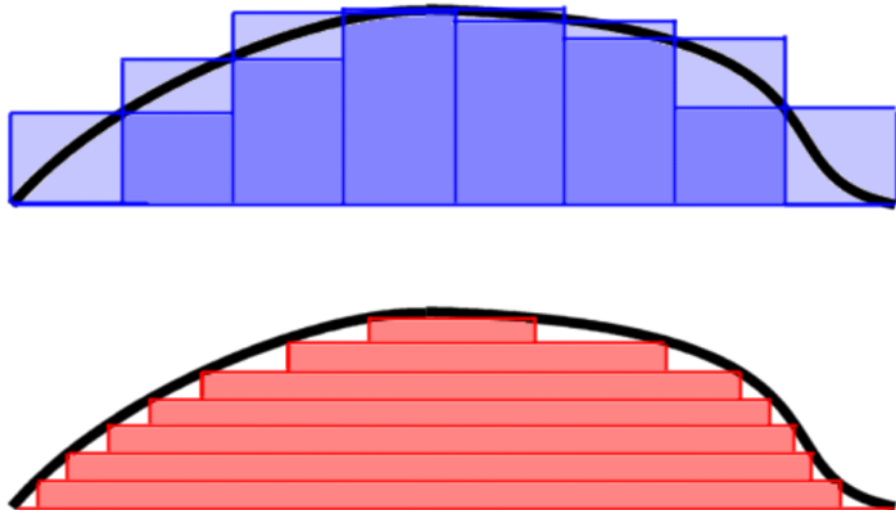
##### Definition

Let  $I = [a, b]$  be an interval in  $\mathbb{R}$  and  $f : I \rightarrow \mathbb{R}$ , be continuous. Partitioning  $I$  using  $n + 1$  nodes  $a = x_0 < x_1 < \dots < x_n = b$  the Riemann sum of  $f$  is defined as

$$\sum_{i=1}^n f(\xi_i)(x_i - x_{i-1}),$$

for  $\xi_i \in [x_{i-1}, x_i]$ . Each choice of  $\xi_i$  defines a different Riemann sum.

#### ii. v.s. Lebesgue Integral



### b. $\sigma$ -algebra

#### i. Definition

Notation  $2^\Omega$  refers to the set of all subsets of  $\Omega$

##### Definition ( $\sigma$ -algebra)

Let  $\Omega$  be a non empty set and  $\mathcal{A} \subset 2^\Omega$  be a class of subsets of  $\Omega$ . Then  $\mathcal{A}$  is a  $\sigma$ -algebra if

i  $\Omega \in \mathcal{A}$ ;

ii  $\mathcal{A}$  is closed under countable unions: if  $A_i \in \mathcal{A}$ , then  $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$ .

### Definition ( $\sigma$ -algebra)

Let  $\Omega$  be a non empty set and  $\mathcal{A} \subset 2^\Omega$  be a class of subsets of  $\Omega$ . Then  $\mathcal{A}$  is a  $\sigma$ -algebra if

- i  $\Omega \in \mathcal{A}$ ;
- ii  $\mathcal{A}$  is *closed under complements*: if  $A \in \mathcal{A}$ , then so is  $\Omega \setminus A$ ;
- iii  $\mathcal{A}$  is *closed under countable unions*: if  $A_0, A_1, \dots$  are in  $\mathcal{A}$ , then so is their union;

#### ii. Remark

This is different from Topology, because Sigma Algebra does not care about open and close.

#### iii. Measure

### Definition

Let  $\Omega$  be a set and  $\mathcal{A}$  be a  $\sigma$ -algebra over  $\Omega$ . A *measure* is a function  $\mu : \mathcal{A} \rightarrow \mathbb{R}_+ \cup \{\infty\}$  which is

- i *Non-negative*: for any  $A \in \mathcal{A}$ ,  $\mu(A) \geq 0$ ;
- ii *Null for the empty set*:  $\mu(\emptyset) = 0$ ;
- iii  *$\sigma$ -additive*: for any countable pairwise disjoint set  $A_i \in \mathcal{A}$ ,  
 $\mu(\bigcup_i A_i) = \sum_{i=0}^{\infty} \mu(A_i)$ ;

### Definitions

- 1 Let  $\Omega$  be a set and  $\mathcal{A}$  be a  $\sigma$ -algebra over  $\Omega$ . Then the pair  $(\Omega, \mathcal{A})$  is called *measurable space*.
- 2 Let  $(\Omega_1, \mathcal{A}_1)$  and  $(\Omega_2, \mathcal{A}_2)$  be two measurable spaces. A function  $f : \Omega_1 \rightarrow \Omega_2$  is measurable if for any  $A \in \mathcal{A}_2$ , the pre-image of  $A$  under  $f$  is in  $\mathcal{A}_1$ .
- 3 A set in a measure space is said to have  *$\sigma$ -finite measure* if it is a countable union of measurable sets with finite measure.
- 4 In a measurable space, a property is said to *hold almost everywhere* if it holds everywhere but on a subset of measure zero.

? "Almost everywhere"

True for everywhere but on a set of measure zero

#### iv. Fubini's Theorem

### Theorem (Fubini)

Let  $(E, X, \mu)$  and  $(F, Y, \nu)$  be two  $\sigma$ -finite measure spaces, and  $(E \times F, X \times Y, \mu\nu)$  be their product measure. If  $f(x, y)$  is  $X \times Y$  integrable, i.e. it is measurable and  $\int_{X \times Y} |f(x, y)| d\mu d\nu$  is finite, then

$$\int_X \left( \int_Y f(x, y) d\nu \right) d\mu = \int_Y \left( \int_X f(x, y) d\mu \right) d\nu = \int_{X \times Y} f(x, y) d\mu d\nu.$$

? Need compact extreme value & intermediate value connected

? Why did I type the above sentence on class

### ★? v. First mean value theorem

We may understand  $g$  as a measure

### Theorem (First mean value theorem)

Let  $f$  be a continuous function over  $[a, b] \subset \mathbb{R}$  and  $g$  is an integrable function which keeps a constant sign on  $[a, b]$ . Then there exists  $c$  in  $(a, b)$  such that

$$\int_a^b f(x)g(x) dx = f(c) \int_a^b g(x) dx.$$

? Why need constant sign

To avoid  $\int g d\mu = 0$  when  $g(x) \neq 0$

c. Taylor's Theorem

i. Theorem

Need absolute continuity

### Theorem (Taylor's theorem)

Given a function  $f$  such that  $f^{(n)}$  is absolutely continuous on the compact  $[a, x] \subset \mathbb{R}$ ,

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{1}{n!} \int_a^x f^{(n+1)}(t) (x-t)^n dt.$$

### ★ii. Weight function

We use weight function to help measure and compare something goes to infinity.

### Definition (Weight function)

A weight function is a continuous function over  $(a, b)$  such that

- i For all  $t \in (a, b)$ ,  $w(t)$  is strictly positive;
- ii The integral  $\int_a^b w(t) dt$  is finite;

## 2. Projection

### a. Definition

We define projection, and we got by-product orthogonality

Works also for functional spaces

### Theorem

Let  $F$  be a vector subspace, of finite dimension, of a normed vector space  $E$ . Then for all  $e \in E$ , there exists a unique element  $u \in F$ , called the *projection of  $e$  on  $F$*  and denoted  $P_F(e)$ , such that

$$\|e - u\| = \min_{f \in F} \|e - f\|.$$

Moreover for any  $f \in F$ ,  $e - P_F(e)$  and  $f$  are orthogonal.

# L18: Orthogonality

2019年7月16日 12:15

## 1. Projection

### a. General Projection

#### i. What's good about continuous function

Continuous function send property between "FROM" and "TO"

#### ii. Projection Theorem: Existence and Uniqueness

$e$  is projected, and  $f$  is its projection

##### Theorem

Let  $F$  be a vector subspace, of finite dimension, of a normed vector space  $E$ . Then for all  $e \in E$ , there exists a unique element  $u \in F$ , called the *projection of  $e$  on  $F$*  and denoted  $P_F(e)$ , such that

$$\|e - u\| = \min_{f \in F} \|e - f\|.$$

Moreover for any  $f \in F$ ,  $e - P_F(e)$  and  $f$  are orthogonal.

### b. Example: Function and Integral

#### i. Definition of norm

Example.

Let  $E$  be  $C[a, b]$  and  $w$  be a weight function over  $E$ . For two functions  $f$  and  $g$  in  $E$ , we define their inner-product in  $E$  by

$$\langle f, g \rangle = \int_a^b f(x)g(x)w(x) dx.$$

#### ii. Definition: Orthogonal bases

##### Definition

An *orthogonal basis* of a normed vector space  $F$  of dimension  $d \in \mathbb{N}$  is a basis of vectors  $\varphi_1, \dots, \varphi_d$  such that

$$\langle \varphi_i, \varphi_j \rangle = \begin{cases} 0 & \text{if } i \neq j \\ a & \text{if } i = j, \end{cases}$$

for some  $a$ . If  $a$  is always 1 then it is called an *orthonormal basis*.

#### iii. Fourier's Coefficient

### Theorem

Let  $(\varphi_1, \dots, \varphi_d)$  be an orthonormal basis for a vector subspace  $F$  of a normed vector space  $E$ . Then for all  $e \in E$ ,

$$P_F(e) = \sum_{j=1}^d \langle e, \varphi_j \rangle \varphi_j.$$

The  $\langle e, \varphi_j \rangle$  are called the *Fourier coefficients* of  $P_F(e)$ .

## iv. Gram-Schmidt Theorem

### Theorem (Gram-Schmidt)

Let  $F$  be a finite dimension inner-product space, with a known basis  $(e_1, \dots, e_d)$ , where  $d$  is the dimension of  $F$ . Then one can construct an orthogonal basis over  $F$ .

## c. Orthogonal Polynomial

### i. Definition

$$\begin{cases} \text{Span}(P_0, \dots, P_n) = \text{Span}(e_0, \dots, e_n) \\ \langle P_i, P_j \rangle = \int_a^b P_i(x) P_j(x) w(x) dx = 0, \quad \text{if } i \neq j \\ P_k(x) = x^k + Q_{k-1}(x), \quad \text{for } Q_{k-1} \in \mathbb{R}_{k-1}[x] \end{cases} \quad (5.3)$$

### Definition

The monic polynomials  $P_i$ ,  $0 \leq i \leq n$ , such that (5.3) is verified are called *orthogonal polynomials*.

### ii. Proposition

? What is topology. Topology look at structure, and find property among and of elements

### Proposition

Let  $P_n$  be the  $(n+1)$ st orthogonal polynomial associated to a weight function  $w$ .

- ① Orthogonal polynomials associated to a weight  $w$  are unique;
- ②  $P_n$  has degree  $n$ ;
- ③  $P_n$  is orthogonal to any polynomial  $Q \in \mathbb{R}_{n-1}[x]$ , for  $n \geq 1$ ;
- ④ For any  $n \geq 1$  there exist  $\lambda_n \in \mathbb{R}$  and  $\mu_n > 0$  such that

$$P_{n+1}(x) = (x - \lambda_n)P_n(x) - \mu_n P_{n-1}(x);$$

- ⑤  $P_n$  has  $n$  distinct real roots in  $(a, b)$ ;

2. Exam: Prove Prove Prove
  - a. Integration, interpolation and \_\_\_\_\_

3. '

# L19: Orthogonality & Interpolation

2019年7月18日 12:20

## 1. Orthogonal Polynomial

### a. Property

#### Proposition

Let  $P_n$  be the  $(n+1)$ st orthogonal polynomial associated to a weight function  $w$ .

- ① Orthogonal polynomials associated to a weight  $w$  are unique;
- ②  $P_n$  has degree  $n$ ;
- ③  $P_n$  is orthogonal to any polynomial  $Q \in \mathbb{R}_{n-1}[x]$ , for  $n \geq 1$ ;
- ④ For any  $n \geq 1$  there exist  $\lambda_n \in \mathbb{R}$  and  $\mu_n > 0$  such that

$$P_{n+1}(x) = (x - \lambda_n)P_n(x) - \mu_n P_{n-1}(x);$$

- ⑤  $P_n$  has  $n$  distinct real roots in  $(a, b)$ ;

### b. Property: Distinct $n$ roots:

Proof: Reduce order, and kill them

### c. Example: Chebyshev Polynomials: $T_n(x) = \cos(n \arccos x)$

#### Proposition

For  $x \in [-1, 1]$ , the function  $T_n(x) = \cos(n \arccos x)$  is called Chebyshev polynomials and verifies the following properties.

- ① For all  $x \in [-1, 1]$ ,  $T_{n+1}(x) + T_{n-1}(x) = 2xT_n(x)$ ;
- ② For all  $n \in \mathbb{N}$ ,  $T_n$  is a polynomial over  $[-1, 1]$ ;
- ③ The  $(T_n)_{n \in \mathbb{N}}$  define a collection of orthogonal polynomials;
- ④ The roots of  $T_n$  are given by  $\cos\left(\frac{2k-1}{2n}\pi\right)$ ,  $1 \leq k \leq n$ ;

## 2. Interpolation

### a. Basic

#### i. What is interpolation

#### ii. Existence and Uniqueness of Polynomial Interpolation

##### Theorem

Let  $f : [a, b] \rightarrow \mathbb{R}$  be a continuous function, and  $x_0, \dots, x_n$  be  $n+1$  distinct points from  $[a, b]$ . There exists a unique polynomial  $P(x) \in \mathbb{R}_n[x]$  such that for all  $0 \leq i \leq n$   $P(x_i) = f(x_i)$ .

### b. Lagrange Polynomial

## ★ i. Lagrange Polynomial

### Definition (Lagrange polynomials)

Let  $f : [a, b] \rightarrow \mathbb{R}$  be a continuous function, and  $x_0, \dots, x_n$  be  $n+1$  distinct points from  $[a, b]$ . The polynomial

$$L_n(f) = \sum_{i=0}^n f(x_i) \ell_i(x), \text{ with } \ell_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}, \text{ for } 0 \leq i \leq n,$$

is called the *Lagrange form of the interpolation polynomial* and the  $\ell_i(x)$ , *Lagrange polynomials*.

## ii. Error of Lagrange Polynomial

### Theorem

Let  $[a, b]$  be a real interval,  $(x_i)_{i=0, \dots, n}$  be  $n+1$  distinct points of  $[a, b]$ , and  $f$  be in  $C^{n+1}[a, b]$ . If  $L_n(f)$  is the Lagrange interpolation polynomial of  $f$  at the points  $x_i$ , then for all  $x$  in  $[a, b]$  there exists  $\xi_x \in [\min_{0 \leq i \leq n}(x, x_i), \max_{0 \leq i \leq n}(x, x_i)] \subset [a, b]$  such that

$$Ef(x) = f(x) - L_n(f)(x) = \frac{f^{(n+1)}(\xi_x)}{(n+1)!} \prod_{j=0}^n (x - x_j).$$

Error will depend on choice of nodes

## iii. Proof of Error of Lagrange Interpolation

### ★ Rolle's theorem

### Theorem (Rolle's theorem)

Let  $f$  be a  $C^n[a, b]$  function, for  $a, b \in \mathbb{R}$ , which has  $n+1$  distinct roots in  $[a, b]$ . Then there exists  $c \in (a, b)$  such that  $f^{(n)}(c) = 0$ .

Construction:

$$Q(t) = P(t) + \frac{f(x) - P(x)}{\prod_n(x)} \prod_n(t)$$

$$g(t) = f(t) - Q(t).$$

## c. Hermite Polynomial

### i. Polynomial

Like Taylor Expansion

### Theorem

Let  $x_0, \dots, x_k$  be  $k+1$  distinct points from  $[a, b]$ , and  $f$  be a continuous function from  $[a, b]$  into  $\mathbb{R}$  such that there is a finite sequence of integers  $\alpha_0, \dots, \alpha_k$  for which  $f^{(l)}(x_i)$  exists,  $0 \leq l \leq \alpha_i$ , and  $i \in \{0, \dots, k\}$ .

Setting  $n = k + \sum_{i=0}^k \alpha_i$ , there exists a unique polynomial  $P$  of degree at most  $n$ , called *Hermite interpolation polynomial*, such that

$$P^{(l)}(x_i) = f^{(l)}(x_i), \quad \text{for } 0 \leq l \leq \alpha_i, \quad i = 0, \dots, k.$$

### ii. Error of Hermite Interpolation

#### Theorem

Let  $x_0, \dots, x_k$  be  $k+1$  distinct points from  $[a, b]$ , and  $f : [a, b] \rightarrow \mathbb{R}$  be in  $C^{n+1}[a, b]$ , such that there is a finite sequence of integers  $\alpha_0, \dots, \alpha_k$  for which  $f^{(l)}(x_i)$  exists,  $0 \leq l \leq \alpha_i$ , and  $i \in \{0, \dots, k\}$ .

If  $P$  is the Hermite interpolation polynomial of  $f$  at  $x_0, \dots, x_k$ , then for all  $x \in [a, b]$  there exists  $\xi_x \in [\min(x, \min_i x_i), \max(x, \max_i x_i)]$  such that

$$f(x) - P(x) = \frac{f^{(n+1)}(\xi_x)}{(n+1)!} \prod_{i=0}^k (x - x_i)^{\alpha_i+1}.$$

### iii. a

a. Error

- ii. We map between space

$$\begin{aligned} \varphi : [-1, 1] &\longrightarrow [a, b] \\ X &\longmapsto \frac{b-a}{2}X + \frac{b+a}{2} \end{aligned}$$

## Theorem

We get exact number from no number.

iv. Mapped error

$$|f(x) - L_n(f)| \leq \frac{2M_n}{(n+1)!} \left( \frac{b-a}{4} \right)^{n+1}.$$

### i. Informal definition

In general, a numerical method is said to *stable* if a small perturbation on the inputs induces a small variation on the result.

## Stability depends on choice of nodes

$$\begin{aligned} L_n(\tilde{f}) - L_n(f) &= \sum_{i=0}^n \tilde{f}(x_i) \ell_i(x) - \sum_{i=0}^n f(x_i) \ell_i(x) \\ &= \sum_{i=0}^n \left( \tilde{f}(x_i) - f(x_i) \right) \ell_i(x) \end{aligned}$$

$$\begin{aligned}
 L_n(\tilde{f}) - L_n(f) &= \sum_{i=0}^n \tilde{f}(x_i) \ell_i(x) - \sum_{i=0}^n f(x_i) \ell_i(x) \\
 &= \sum_{i=0}^n (\tilde{f}(x_i) - f(x_i)) \ell_i(x) \\
 &= L_n(\tilde{f} - f),
 \end{aligned}$$

$$|L_n(\varepsilon)| \leq \max |\tilde{f}(x_i) - f(x_i)| \cdot \max_{x \in [a,b]} \sum_{i=0}^n |\ell_i(x)|.$$

We name it Lebesgue constant

### Definition

In Lagrange interpolation the number  $\Lambda_n = \max_{x \in [a,b]} \sum_{i=0}^n |\ell_i(x)|$  is called the *Lebesgue constant*.

iii. '

### Theorem

Let  $E = C[a, b]$  for  $[a, b]$  a real interval and  $F = \mathbb{R}_n[x]$ . Then  $L_n$  is in  $\mathcal{L}(E, F)$  and  $\|L_n\| = \Lambda_n$ .

What trick is used in this proof

2.

# L21: Gauss's Method

2019年7月25日 12:19

## 1. Gauss's Method

### a. Introduction

We can interpolate function into polynomial, and then integrate those polynomial

### b. Initial problem

#### i. Condition

? **compact**

#### ii. Lemma

Integration is also a linear function

#### iii. Definition: Order of method

### Definitions

Let  $(I_k(f))_{k \geq 0}$  be a sequence of approximation of  $I(f)$ .

- ① The sequence  $I_k(f)$  is *convergent* if  $\lim_{k \rightarrow \infty} I_k(f) = I(f)$ ;
- ② The *error of approximation* is given by  $E_k(f) = I(f) - I_k(f)$ ;
- ③ An integration method is said to be of *order*  $N$ ,  $N \in \mathbb{N}$ , if for any polynomial  $P \in \mathbb{R}_N[x]$ ,  $E_k(P) = 0$ ,  $k \geq N$ ;
- ④ An integration method is said to be of *order exactly*  $N$  if it is of order  $N$  and there exists  $P \in \mathbb{R}_{N+1}[x]$  such that  $E_k(P) \neq 0$  for some  $k \geq N$ ;

Read the example 7.1

### c. **Existence and Uniqueness of Interpolation**

#### i. Theorem

Use weight function to choose nodes, and then use those nodes to interpolate

### Theorem

Let  $k$  and  $l$  be two integers such that  $k \geq l - 1$  and  $x_0, \dots, x_{l-1}$  be in  $[\alpha, \beta]$  while  $x_l, \dots, x_k$  are not in this interval. For  $x$  in  $(\alpha, \beta)$ , let  $\Pi_{k,l}(x)$  denote the product  $\prod_{j=l}^k (x - x_j)$ . Then there exists a quadrature formula that is uniquely determined by  $x_l, \dots, x_k$ . It is of the form

$$\sum_{i=0}^k \lambda_i f(x_i) \approx \int_{\alpha}^{\beta} f(x) w(x) dx$$

and has order  $k + l$ . Moreover the  $x_i$ ,  $0 \leq i \leq l - 1$ , are the roots of the  $(l + 1)$ th polynomial  $P_l$  associated to the weight

$$\theta(x) = \text{sign}(\Pi_{k,l}(x)) \Pi_{k,l}(x) w(x), \quad x \in (\alpha, \beta).$$

#### ii. Corollary

Choose all nodes inside

### Corollary

When  $k = l - 1$ , i.e. no node is taken out of  $(\alpha, \beta)$ , there exists a unique quadrature formula of the form

$$\int_{\alpha}^{\beta} f(x)w(x) dx \approx \sum_{i=0}^k \lambda_i f(x_i),$$

and with order  $2k + 1$ . The  $x_i \in (\alpha, \beta)$ ,  $0 \leq i \leq k$ , are the roots of the  $(k + 2)$ nd orthogonal polynomial associated to the weight  $w$ . Moreover all the  $\lambda_i$  are strictly positive.

#### d. Gauss's Method

##### i. Idea

##### ii. Theorem

Choose outside nodes, then inner nodes will be fixed by the root of orthogonal polynomial.

Prove: very important

1. '

a. \_\_\_\_\_

i. Proof on 170

Extremely important

ii. ;

When plugging function, we may choose  $x^k$  or Lagrangian polynomial

? ★ iii. Hermite Interpolation

**Theorem**

Let  $x_0, \dots, x_k$  be  $k+1$  distinct points from  $[a, b]$ , and  $f$  be a continuous function from  $[a, b]$  into  $\mathbb{R}$  such that there is a finite sequence of integers  $\alpha_0, \dots, \alpha_k$  for which  $f^{(l)}(x_i)$  exists,  $0 \leq l \leq \alpha_i$ , and  $i \in \{0, \dots, k\}$ .

Setting  $n = k + \sum_{i=0}^k \alpha_i$ , there exists a unique polynomial  $P$  of degree at most  $n$ , called *Hermite interpolation polynomial*, such that

$$P^{(l)}(x_i) = f^{(l)}(x_i), \quad \text{for } 0 \leq l \leq \alpha_i, \quad i = 0, \dots, k.$$

Proof

**Definition**

Let  $E$  and  $F$  be two vector spaces, and  $f$  be a linear application from  $E$  to  $F$ . The *kernel* of  $f$ , noted  $\ker f$ , is the set of all the elements  $x$  from  $E$  such that  $f(x) = 0$ .

**Lemma**

For two vector spaces  $E$  and  $F$  of similar dimension, a linear application  $f$  from  $E$  into  $F$  is bijective if and only if  $\ker f = \{0\}$ .

Error

### Theorem

Let  $x_0, \dots, x_k$  be  $k+1$  distinct points from  $[a, b]$ , and  $f : [a, b] \rightarrow \mathbb{R}$  be in  $C^{n+1}[a, b]$ , such that there is a finite sequence of integers  $\alpha_0, \dots, \alpha_k$  for which  $f^{(l)}(x_i)$  exists,  $0 \leq l \leq \alpha_i$ , and  $i \in \{0, \dots, k\}$ . If  $P$  is the Hermite interpolation polynomial of  $f$  at  $x_0, \dots, x_k$ , then for all  $x \in [a, b]$  there exists  $\xi_x \in [\min(x, \min_i x_i), \max(x, \max_i x_i)]$  such that

$$f(x) - P(x) = \frac{f^{(n+1)}(\xi_x)}{(n+1)!} \prod_{i=0}^k (x - x_i)^{\alpha_i+1}.$$

#### iv. Error of Gauss's

Use Hermite result to prove

### Theorem

Let  $x_0, \dots, x_{l-1}$  be the nodes from  $(\alpha, \beta)$  and  $x_l, \dots, x_k$  be the points of  $[a, b]$  exterior to  $(\alpha, \beta)$ . Then for any function  $f \in C^{k+l+1}[a, b]$  there exists  $\xi$  in  $[a, b]$  such that

$$E(f) = \frac{f^{(k+l+1)}(\xi)}{(k+l+1)!} \int_{\alpha}^{\beta} \prod_{i=0}^{l-1} (x - x_i)^2 \prod_{j=l}^k (x - x_j) w(x) dx.$$

#### v. Exact order

### Corollary

Gauss' method has order exactly  $k + l$ .

#### b. Peano

##### ★ i. What is it

Let  $w(x)$  be a weight function over  $(\alpha, \beta) \subset [a, b]$  and  $f$  be a function defined almost everywhere on  $[a, b]$  and which is Lebesgue integrable on  $(\alpha, \beta)$  with respect to the weight  $w$ . We are interested in the generic quadrature formula

$$\int_{\alpha}^{\beta} f(x) w(x) dx \approx \sum_{i=0}^k \lambda_i f(x_i), \quad \lambda_i \in \mathbb{R}, x_i \in [a, b], \forall 0 \leq i \leq k.$$

This method is called *Peano's method*.

##### ii. Peano's Kernel

### Definition

The application  $K_N$ , called *Peano kernel*, is defined by

$$K_N(t) = E(x \in [\alpha, \beta] \mapsto [(x - t)_+]^N),$$

where

$$[(x - t)_+]^N = \begin{cases} (x - t)^N, & \text{if } t \leq x \\ 0, & \text{if } t > x \end{cases}$$

### Theorem (Peano kernel theorem)

Assuming Peano's method is of positive order  $N$ ,  $f$  is a  $C^{N+1}[a, b]$  function, and  $f^{(N)}$  is absolutely continuous then

$$E(f) = \frac{1}{N!} \int_a^b K_N(t) f^{(N+1)}(t) dt.$$

### Corollary

Under the same assumptions as Peano kernel theorem we have

$$|E(f)| \leq \frac{1}{N!} \sup_{t \in [a, b]} |f^{(N+1)}(t)| \int_a^b |K_N(t)| dt.$$

90

## Peano kernel theorem

### Corollary

Under the same assumptions as Peano kernel theorem and assuming Peano kernel does not change sign, there exists  $\xi \in (a, b)$  such that

$$E(f) = \frac{f^{(N+1)}(\xi)}{(N+1)!} E(x \mapsto x^{N+1}).$$

Beautiful proof: plug in a known function, and get the value for general function

2. '

## L23:

2019年7月31日 14:01

1. '

a. What is  $E(x \rightarrow x^n)$

b. What is  $\lambda_{jk}$

c. Polya's convergence

Very generic. We don't require choice of  $\lambda_{jk}$ , (the choice of nodes).

By i, what are we requiring?3

We have defined the method, thus knowing choice of nodes

### Theorem (Polya's convergence)

Using the previous notations, the error  $E_k(f)$  converges to 0 as  $k$  tends to infinity if and only if the following two conditions are met

- i For any positive integer  $n$ ,  $\lim_{k \rightarrow \infty} E_k(x \mapsto x^n) = 0$ ;
- ii There exists  $M$ , strictly positive, such that for all  $k \in \mathbb{N}$   
$$\sum_{j=0}^k |\lambda_{jk}| \leq M;$$

Norm

2. '

1. '

- a. Why Continuous function important in topology  
Topology structure is preserved by continuous function  
Transfer
- b. Stone-W, why is it important

**Theorem (Stone-Weierstrass)**

Let  $X$  be a compact metric space and  $\mathcal{H}$  be a subalgebra that separates  $C(X, \mathbb{R})$ . Then any element from  $C(X, \mathbb{R})$  is the uniform limit of a sequence of elements from  $\mathcal{H}$ .

- c. Rolle theorem  
How we use it.  
Where Rolle is necessary
- d. Hermite interpolation  
Interpolate based on both  $f$  and  $f'$
- e. Polya convergence  
Why important, why powerful

2. '

- a. When prove Gauss', think in terms of Peano's method  
,
- b. Orthogonal polynomial  
"
- c. Weight function  
Need to prove what is a weight function
- d. \_\_\_\_\_  
,

3. Range

- a. Ortho, interpolation, integration. (we need topology understanding, but don't need to prove topology: compact, norm)
- b. What is this exercise about. What is the big picture (15')



Exam

开始录音: 2019年7月31日 15:18

## 1. Topology Basics

### a. Norm

- ② The map  $\|\cdot\| : V \rightarrow \mathbb{R}_+$  is called a **norm** if for any  $u, v \in V$  and  $a \in \mathbb{K}$  it satisfies:

- ①  $\|u\| = 0 \Leftrightarrow u = 0$ ;
- ②  $\|au\| = |a|\|u\|$ ;
- ③ **Triangular inequality**:  $\|u + v\| \leq \|u\| + \|v\|$ ;

A vector space with an **norm** is an **normed vector space**.

### b. Continuity

#### Theorem

Let  $E$  and  $F$  be two normed vector spaces over  $\mathbb{K}$  and  $u$  be a linear application. The following conditions are equivalent.

- i  $u \in \mathcal{L}(E, F)$ ;
- ii  $u$  is uniformly continuous;
- iii  $u$  is continuous;
- iv  $u$  is continuous for any  $x \in E$ ;
- v  $u$  is bounded over any subset of  $E$ ;
- vi There exists  $a \in E$  and  $r \in \mathbb{R}_+^*$  such that  $u$  is bounded over  $B(a, r)$ ;

### c. Continuous map

[Only] close/open property from "TO" to "FROM"  
compact from "FROM" to "TO"

**Intermediate Value Theorem**: need connectness

**Extreme Value Theorem**: need compact

### ★ d. Banach-Steinhaus Theorem (Very strong, iff)

#### Theorem (Banach-Steinhaus)

Let  $E$  be a Banach space,  $F$  be a normed vector space over  $\mathbb{K}$ , and  $H$  be a non-empty subset of  $\mathcal{L}(E, F)$ . Then  $\sup\{\|u\| : u \in H\}$  is finite if and only if for all  $x \in E$ ,  $\sup\{\|u(x)\| : u \in H\}$  is also finite.

### e. Stone-Weierstrass Theorem

#### Theorem (Stone-Weierstrass)

Let  $X$  be a compact metric space and  $\mathcal{H}$  be a subalgebra that separates  $C(X, \mathbb{R})$ . Then any element from  $C(X, \mathbb{R})$  is the uniform limit of a sequence of elements from  $\mathcal{H}$ .

## 2. Integral

### a. Weight function

$$w(t) > 0, \left| \int_a^b w(t) dt \right| < \infty$$

### ★ b. First mean value theorem

#### Theorem (First mean value theorem)

Let  $f$  be a continuous function over  $[a, b] \subset \mathbb{R}$  and  $g$  is an integrable function which keeps a constant sign on  $[a, b]$ . Then there exists  $c$  in  $(a, b)$  such that

$$\int_a^b f(x)g(x) dx = f(c) \int_a^b g(x) dx.$$

### c. Taylor's Theorem

Need **absolute continuity**

#### Theorem (Taylor's theorem)

Given a function  $f$  such that  $f^{(n)}$  is absolutely continuous on the compact  $[a, x] \subset \mathbb{R}$ ,

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{1}{n!} \int_a^x f^{(n+1)}(t) (x-t)^n dt.$$

## 3. Orthogonality

### a. Inner product

$$\langle f, g \rangle = \int_a^b f(x)g(x)w(x) dx.$$

### b. Gram-Schmidt Theorem: Possibility of Normalization

**Theorem (Gram-Schmidt)**

Let  $F$  be a finite dimension inner-product space, with a known basis  $(e_1, \dots, e_d)$ , where  $d$  is the dimension of  $F$ . Then one can construct an orthogonal basis over  $F$ .

**c. Property of Orthogonal Polynomial****Proposition**

Let  $P_n$  be the  $(n+1)$ st orthogonal polynomial associated to a weight function  $w$ .

- ① Orthogonal polynomials associated to a weight  $w$  are unique;
- ②  $P_n$  has degree  $n$ ;
- ③  $P_n$  is orthogonal to any polynomial  $Q \in \mathbb{R}_{n-1}[x]$ , for  $n \geq 1$ ;
- ④ For any  $n \geq 1$  there exist  $\lambda_n \in \mathbb{R}$  and  $\mu_n > 0$  such that

$$P_{n+1}(x) = (x - \lambda_n)P_n(x) - \mu_n P_{n-1}(x);$$

- ⑤  $P_n$  has  $n$  distinct real roots in  $(a, b)$ ;

**d. Chebyshev, a wise choice****Proposition**

For  $x \in [-1, 1]$ , the function  $T_n(x) = \cos(n \arccos x)$  is called Chebyshev polynomials and verifies the following properties.

- ① For all  $x \in [-1, 1]$ ,  $T_{n+1}(x) + T_{n-1}(x) = 2xT_n(x)$ ;
- ② For all  $n \in \mathbb{N}$ ,  $T_n$  is a polynomial over  $[-1, 1]$ ;
- ③ The  $(T_n)_{n \in \mathbb{N}}$  define a collection of orthogonal polynomials;
- ④ The roots of  $T_n$  are given by  $\cos\left(\frac{2k-1}{2n}\pi\right)$ ,  $1 \leq k \leq n$ ;

**4. Interpolation****a. Cases**

|                | Lagrange Polynomial                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Hermite Polynomial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition     | <b>Definition (Lagrange polynomials)</b><br>Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function, and $x_0, \dots, x_n$ be $n+1$ distinct points from $[a, b]$ . The polynomial $L_n(f) = \sum_{i=0}^n f(x_i) \ell_i(x), \text{ with } \ell_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}, \text{ for } 0 \leq i \leq n,$ is called the <i>Lagrange form of the interpolation polynomial</i> and the $\ell_i(x)$ , <i>Lagrange polynomials</i> . | <b>Theorem</b><br>Let $x_0, \dots, x_k$ be $k+1$ distinct points from $[a, b]$ , and $f$ be a continuous function from $[a, b]$ into $\mathbb{R}$ such that there is a finite sequence of integers $\alpha_0, \dots, \alpha_k$ for which $f^{(l)}(x_i)$ exists, $0 \leq l \leq \alpha_i$ , and $i \in \{0, \dots, k\}$ .<br>Setting $n = k + \sum_{i=0}^k \alpha_i$ , there exists a unique polynomial $P$ of degree at most $n$ , called <i>Hermite interpolation polynomial</i> , such that $P^{(l)}(x_i) = f^{(l)}(x_i), \text{ for } 0 \leq l \leq \alpha_i, i = 0, \dots, k.$ |
| Error          | $\xi_x \in [\min(x, \min_{0 \leq i \leq n} x_i), \max(x, \max_{0 \leq i \leq n} x_i)] \subset [a, b]$ such that $Ef(x) = f(x) - L_n(f)(x) = \frac{f^{(n+1)}(\xi_x)}{(n+1)!} \prod_{j=0}^n (x - x_j).$                                                                                                                                                                                                                                                                                | for all $x \in [a, b]$ there exists $\xi_x \in [\min(x, \min_{i=0, \dots, k} x_i), \max(x, \max_{i=0, \dots, k} x_i)]$ such that $f(x) - P(x) = \frac{f^{(n+1)}(\xi_x)}{(n+1)!} \prod_{i=0}^k (x - x_i)^{\alpha_i + 1}.$                                                                                                                                                                                                                                                                                                                                                           |
| Proof of error | <b>Theorem (Rolle's theorem)</b><br>Let $f$ be a $C^n[a, b]$ function, for $a, b \in \mathbb{R}$ , which has $n+1$ distinct roots in $[a, b]$ . Then there exists $c \in (a, b)$ such that $f^{(n)}(c) = 0$ .<br>$Q(t) = P(t) + \frac{f(x) - P(x)}{\prod_{i=0}^n (x_i - x)} \prod_{i=0}^n (t - x_i)$ $g(t) = f(t) - Q(t).$                                                                                                                                                           | <b>Definition</b><br>Let $E$ and $F$ be two vector spaces, and $f$ be a linear application from $E$ to $F$ . The <i>kernel</i> of $f$ , noted $\ker f$ , is the set of all the elements $x$ from $E$ such that $f(x) = 0$ .<br><b>Lemma</b><br>For two vector spaces $E$ and $F$ of similar dimension, a linear application $f$ from $E$ into $F$ is bijective if and only if $\ker f = \{0\}$ .                                                                                                                                                                                   |

**b. General Error****Theorem**

For any polynomial  $q$  in  $E_n$ ,  $\max_{t \in [-1, 1]} |q(t)| \geq 2^{1-n}$ .

We get exact number from no number.

Chebyshev touch this minimum error

**5. Integration****a. Generic: Peano****i. Method**

$f$  defined almost everywhere

$$\int_{\alpha}^{\beta} f(x) w(x) dx \approx \sum_{i=0}^k \lambda_i f(x_i), \quad \lambda_i \in \mathbb{R}, x_i \in [a, b], \forall 0 \leq i \leq k.$$

**ii. Error: Peano Kernel Theorem**

### Definition

The application  $K_N$ , called *Peano kernel*, is defined by

$$K_N(t) = E(x \in [\alpha, \beta] \mapsto [(x - t)_+]^N),$$

where

$$[(x - t)_+]^N = \begin{cases} (x - t)^N, & \text{if } t \leq x \\ 0, & \text{if } t > x \end{cases}$$

### Theorem (Peano kernel theorem)

Assuming Peano's method is of positive order  $N$ ,  $f$  is a  $C^{N+1}[a, b]$  function, and  $f^{(N)}$  is absolutely continuous then

$$E(f) = \frac{1}{N!} \int_a^b K_N(t) f^{(N+1)}(t) dt.$$

iii. Corollary:  $K_N(t)$  doesn't change sign

$$E(f) = \frac{f^{(N+1)}(\xi)}{(N+1)!} E(x \mapsto x^{N+1}).$$

iv. Theorem: Polya's Convergence

### Theorem (Polya's convergence)

Using the previous notations, the error  $E_k(f)$  converges to 0 as  $k$  tends to infinity if and only if the following two conditions are met

- i For any positive integer  $n$ ,  $\lim_{k \rightarrow \infty} E_k(x \mapsto x^n) = 0$ ;
- ii There exists  $M$ , strictly positive, such that for all  $k \in \mathbb{N}$   $\sum_{j=0}^k |\lambda_{jk}| \leq M$ ;

b. Cases

| / Method | Integration through interpolation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Gauss's Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | <p><b>Theorem</b></p> <p>Let <math>f</math> be a continuous function over <math>[\alpha, \beta] \subset [a, b]</math>. Using the previous notations, let <math>(l_k)_{k \in \mathbb{N}}</math> be the sequence defined by</p> $l_k = \sum_{i=0}^{k-1} h_i \sum_{j=0}^l f(\alpha_{ij}) w_j.$ <p>Then <math>(l_k)_{k \in \mathbb{N}}</math> converges toward <math>\int_{\alpha}^{\beta} f(x) dx</math> as <math>k</math> tends to infinity. Moreover the method of integration through interpolation over <math>l+1</math> nodes has order at least <math>l</math>.</p> | <p><b>Theorem</b></p> <p>Let <math>k</math> and <math>l</math> be two integers such that <math>k \geq l-1</math> and <math>x_0, \dots, x_{l-1}</math> be in <math>[\alpha, \beta]</math> while <math>x_l, \dots, x_k</math> are not in this interval. For <math>x</math> in <math>(\alpha, \beta)</math>, let <math>\Pi_{k,l}(x)</math> denote the product <math>\prod_{j=l}^k (x - x_j)</math>. Then there exists a quadrature formula that is uniquely determined by <math>x_l, \dots, x_k</math>. It is of the form</p> $\sum_{i=0}^k \lambda_i f(x_i) \approx \int_{\alpha}^{\beta} f(x) w(x) dx$ <p>and has order <math>k+l</math>. Moreover the <math>x_i</math>, <math>0 \leq i \leq l-1</math>, are the roots of the <math>(l+1)</math>th polynomial <math>P_l</math> associated to the weight</p> $\theta(x) = \text{sign}(\Pi_{k,l}(x)) \Pi_{k,l}(x) w(x), \quad x \in (\alpha, \beta).$ <p>If all nodes are inside, then all <math>\lambda_i &gt; 0</math><br/> <b>Write exact order</b></p> |
| Error    | <p><b>Proposition</b></p> <p>If <math>f \in C^{N+1}[\alpha, \beta]</math> and the method has order <math>N</math>, then</p> $ E_{\text{comp}}(f)  \leq \frac{C_{N,f}}{2^{N+2}} (\beta - \alpha) h^{N+1},$ <p>where</p> $\begin{cases} h = \max_{0 \leq i \leq k-1} h_i \\ C_{N,f} = \frac{\max_{\alpha, \beta}  f^{(N+1)} }{N!} \int_{-1}^1  k_N(t)  dt. \end{cases}$                                                                                                                                                                                                  | <p><b>Theorem</b></p> <p>Let <math>x_0, \dots, x_{l-1}</math> be the nodes from <math>(\alpha, \beta)</math> and <math>x_l, \dots, x_k</math> be the points of <math>[a, b]</math> exterior to <math>(\alpha, \beta)</math>. Then for any function <math>f \in C^{k+l+1}[a, b]</math> there exists <math>\xi</math> in <math>[a, b]</math> such that</p> $E(f) = \frac{f^{(k+l+1)}(\xi)}{(k+l+1)!} \int_{\alpha}^{\beta} \prod_{i=0}^{l-1} (x - x_i)^2 \prod_{j=l}^k (x - x_j) w(x) dx.$ <p><math>K_N(t)</math> keeps const sign</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

c.